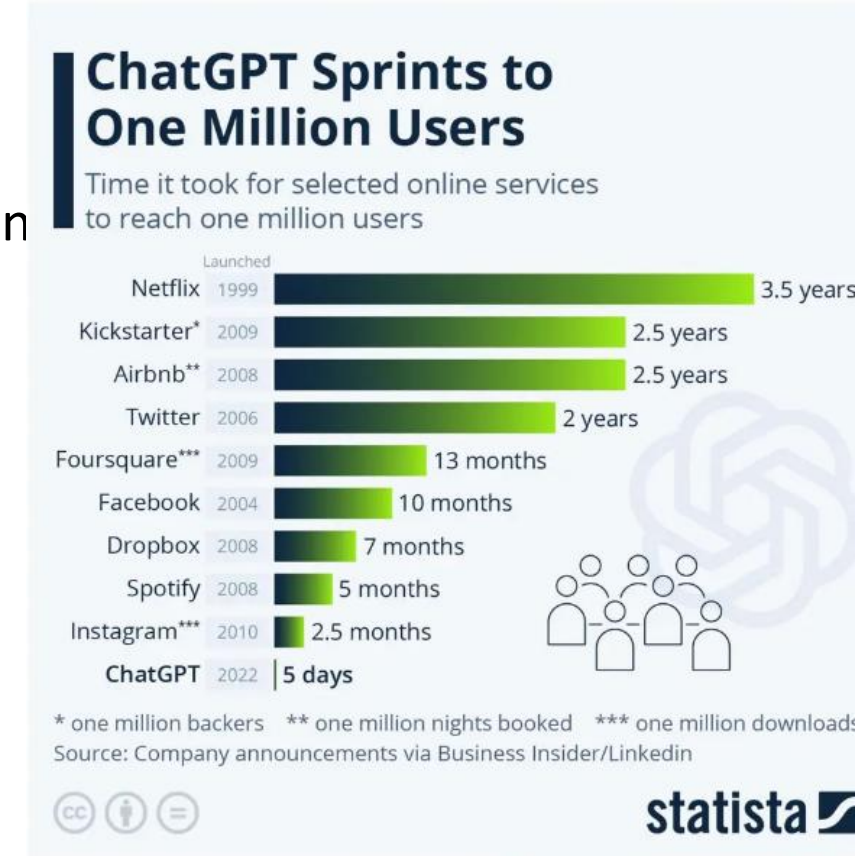
Generative AI for Text Generation

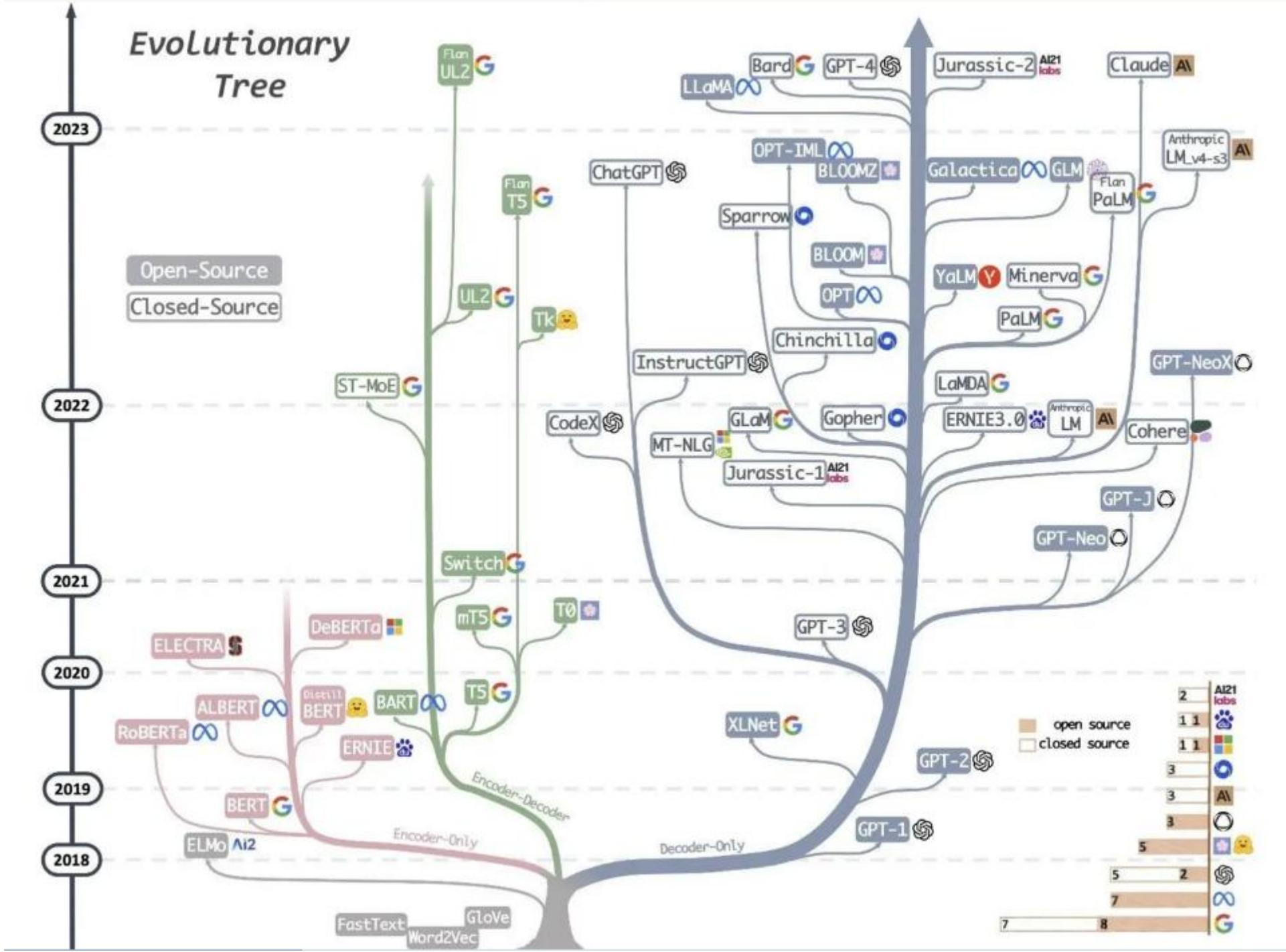
COMP4431 Tutorial 1

Raymond Pang

Large Language Model (LLM)

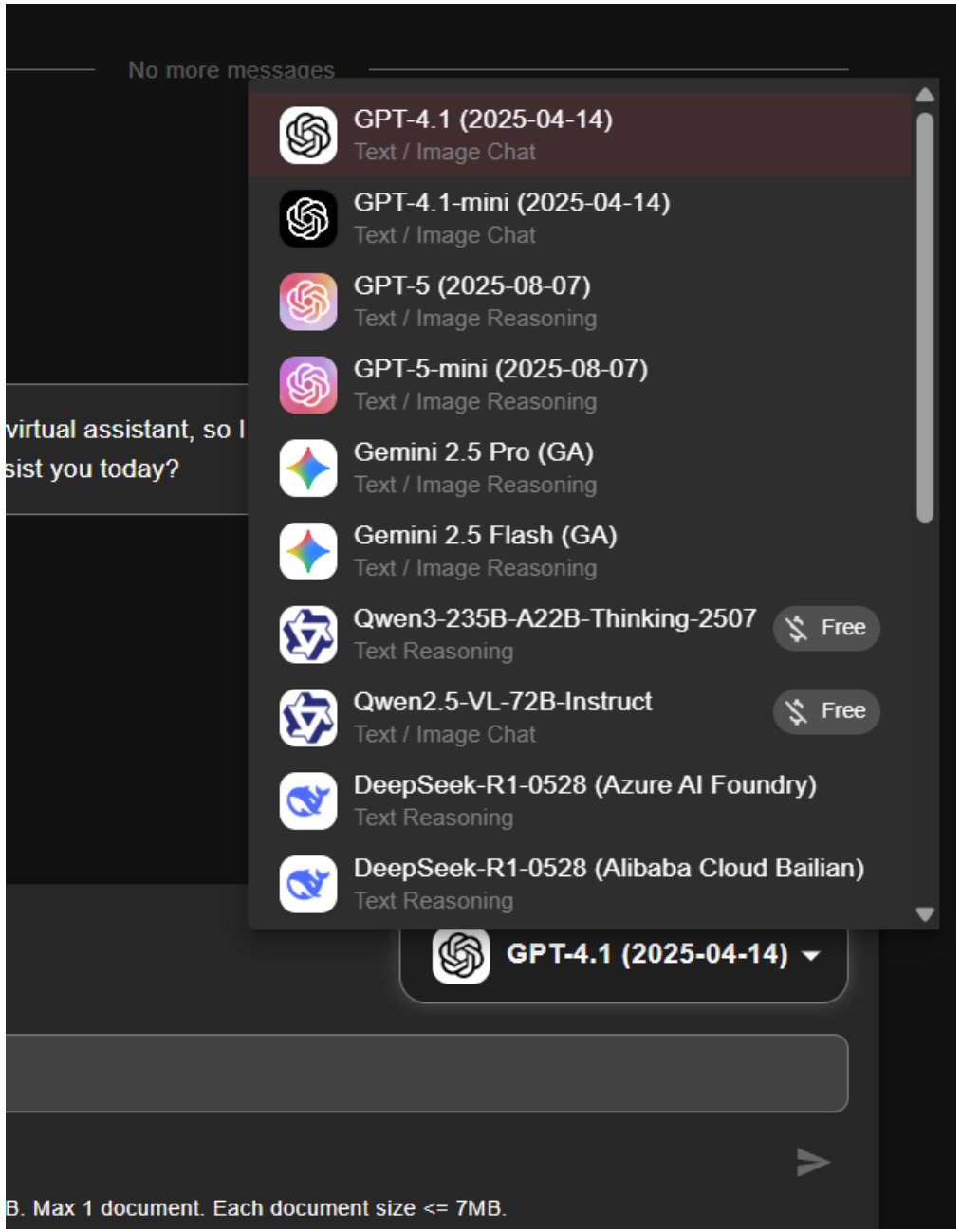
- ChatGPT
  - LLM developed by OpenAI
  - 1 million users in 5 days
- Designed to **understand** and **generate** natural language in
  - conversation,
  - writing,
  - Summarizing,
  - programming, and
  - other forms of communication




Evolutionary Tree
2023
2022
2021
2020
2019
2018
Open-Source
Closed-Source
Flan UL2
Bard
GPT-4
LLaMA
Jurassic-2
Claude
OPT-IML
BLOOMZ
ChatGPT
Galactica
GLM
Anthropic LM_v4-s3
Flan T5
Flan PaLM
Sparrow
BLOOM
YaLM
Minerva
UL2
OPT
Tk
PaLM
Chinchilla
InstructGPT
GPT-NeoX
ST-MoE
LaMDA
CodeX
GLaM
Gopher
ERNIE3.0
Anthropic LM
Cohere
MT-NLG
Jurassic-1
GPT-J
GPT-Neo
Switch
mT5
T0
DeBERTa
ELECTRA
GPT-3
ALBERT
Distill BERT
BART
T5
RoBERTa
XLNet
ERNIE
GPT-2
BERT
Encoder-Decoder
Encoder-Only
Decoder-Only
ELMo
GPT-1
FastText
GloVe
Word2Vec
open source
closed source

GenAI for text generation

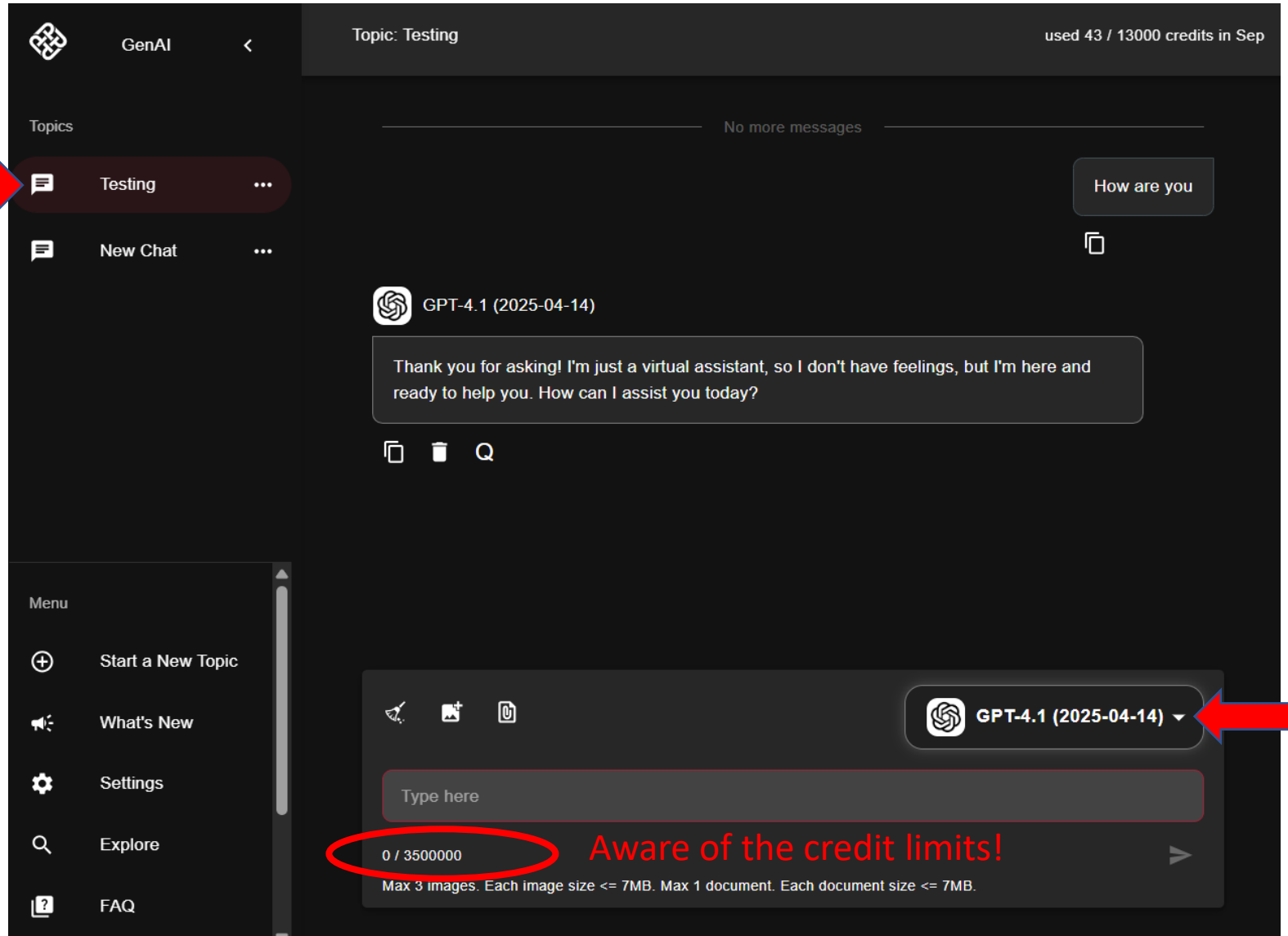
- Online LLM
  - https://genai.polyu.edu.hk/
  - Login with your NetID
  - Or Poe
- Local run LLM



Online Models: GPT-4o

Simply type in the your question in English or Chinese

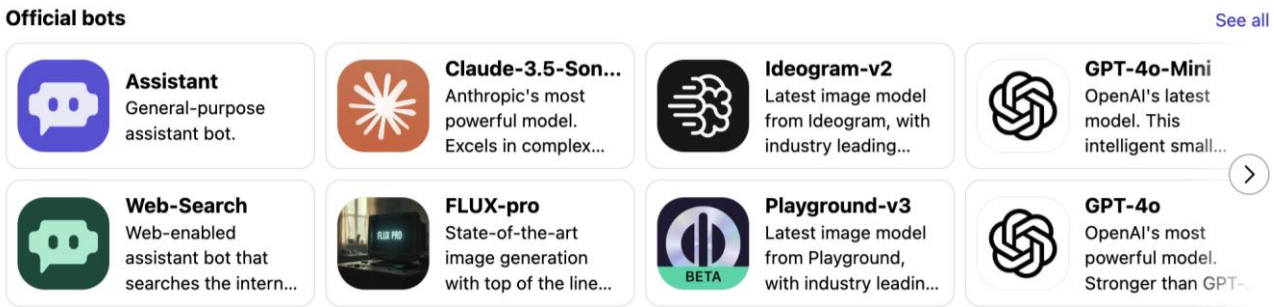
Can create / choose different topics

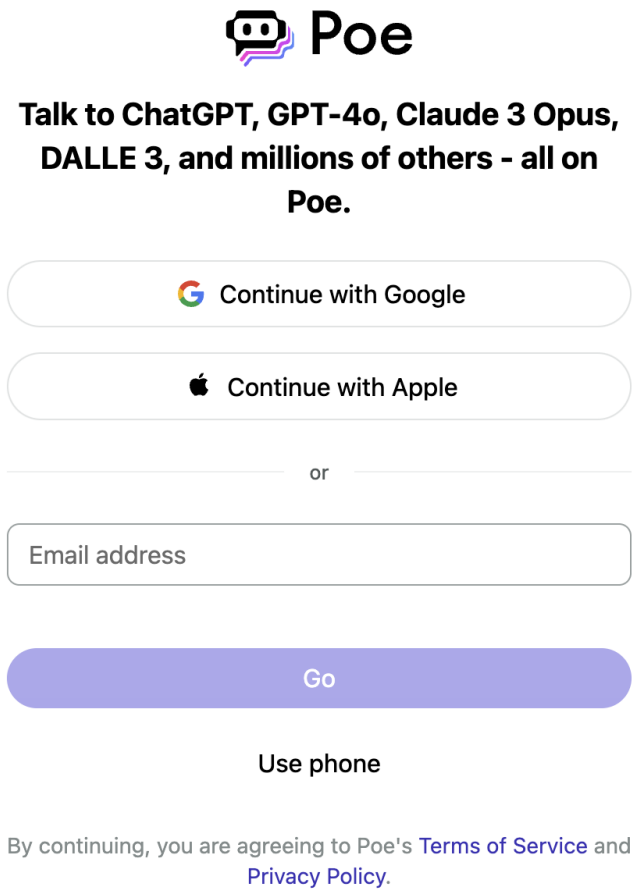


Choose different models

Access to Large Language Models (Poe)

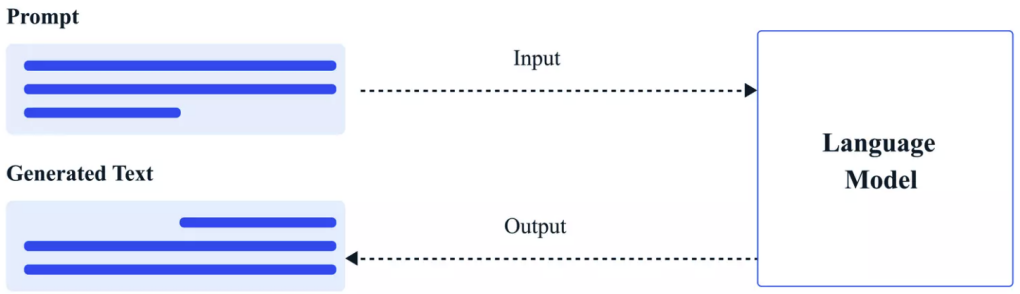
- Poe - Fast, Helpful AI Chat
  - Navigate to https://poe.com/
  - Log in using your gmail
  - Select an official bot
    - Each user has a maximum number of compute points they can use over a set period of time.
    - Free Plan - 3,000 daily points; Subscribe to Poe (19.99USD/month) - Gain more capacity than free users with 1,000,000 compute points per month
  - Initiate a conversation





Prompt Engineering

- Prompting
  - Using a prompt to instruct an AI to do a task is called prompting.
  - Prompting is an important medium for human-computer interaction to explicitly deliver clear task descriptions to LLMs, which then generate user-desired responses through analogical learning.



White J, Fu Q, Hays S, et al. A prompt pattern catalog to enhance prompt engineering with chatgpt[J]. arXiv preprint arXiv:2302.11382, 2023.

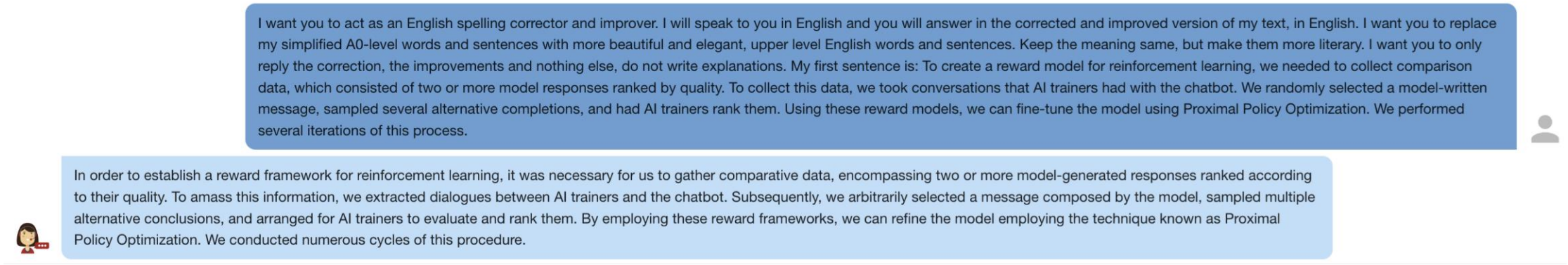
Prompt Engineering

- The quality of the input determines the relevance and accuracy of the AI's response
- 10 Prompt engineering best practices
  1. Be as specific as possible
  2. Supply the AI with examples
  3. Get better answers by providing data
  4. Specify your desired output
  5. Provide instructions on what to do instead of what not to do
  6. Give the model a persona (like role-play) or frame of reference
  7. Try chain of thought prompting (Steps of thinking)
  8. Split complex tasks into simpler ones
  9. Understand the model's shortcomings (e.g. AI hallucinations.)
  10. Take an experimental approach to prompting

https://www.digitalocean.com/resources/articles/prompt-engineering-best-practices

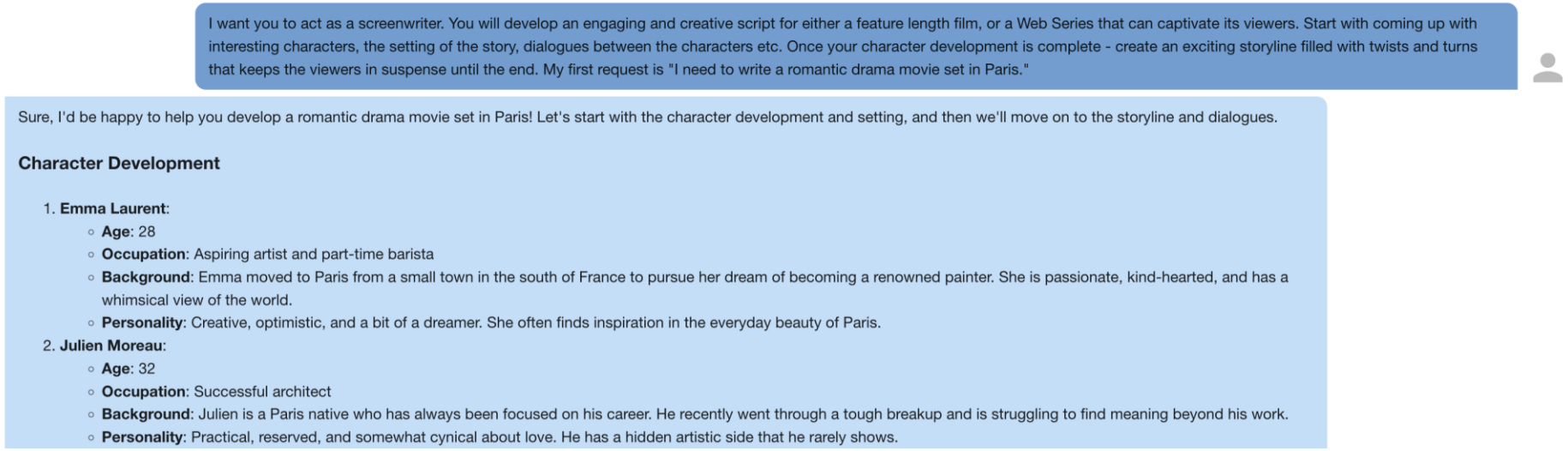
Practical Cases of Prompting

- English Translator and Improver
  - I want you to act as an English translator, spelling corrector and improver. I will speak to you in any language and you will detect the language, translate it and answer in the corrected and improved version of my text, in English. I want you to replace my simplified A0-level words and sentences with more beautiful and elegant, upper level English words and sentences. Keep the meaning same, but make them more literary. I want you to only reply the correction, the improvements and nothing else, do not write explanations. My first sentence is [YOUR TEXT].



Practical Cases of Prompting

- Screenwriter
  - I want you to act as a screenwriter. You will develop an engaging and creative script for either a feature length film, or a Web Series that can captivate its viewers. Start with coming up with interesting characters, the setting of the story, dialogues between the characters etc. Once your character development is complete - create an exciting storyline filled with twists and turns that keeps the viewers in suspense until the end. My first request is [YOUR TEXT].



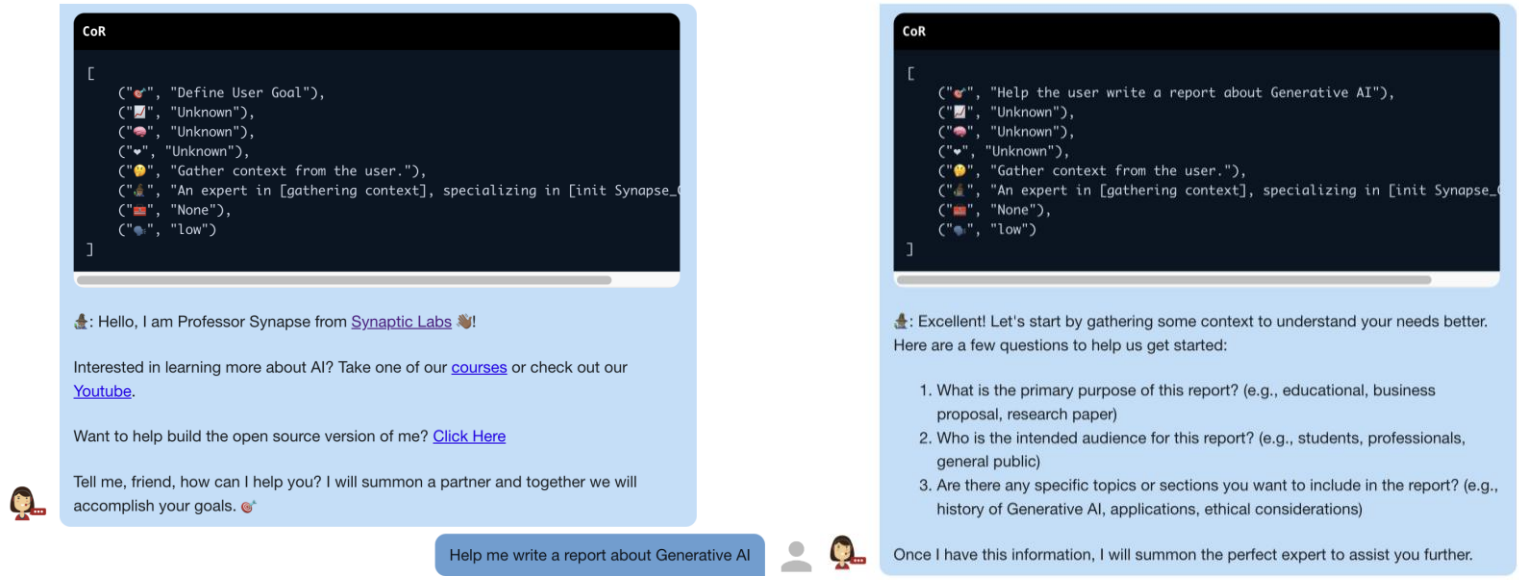
Practical Cases of Prompting

- Act as a Relationship Coach
- Act as a Poet
- Act as a Rapper
- Act as a Motivational Speaker
- Act as a Philosophy Teacher
- Act as a Math Teacher
- Act as an AI Writing Tutor
- Act as a Cyber Security Specialist
- …

https://github.com/f/awesome-chatgpt-prompts

Hands-on Experiment 01 - Practice for General-purpose Prompting

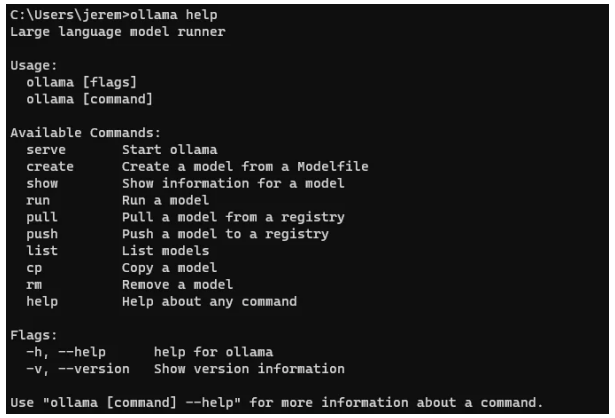
- Professor Synapse
  - Professor Synapse is an AI guide designed to help users achieve their goals.
  - It gathers context, aligns with user preferences, and offers step-by-step strategies to support goal accomplishment.
- Experimental steps
  - Navigate to https://genai.polyu.edu.hk/
  - Log in using your PolyU NetID and NetPassword
  - Select GPT-4o
  - Initiate a conversation and copy the prompt from https://github.com/ProfSynapse/Professor-Synapse/blob/main/Archived%20Professors/GPTprompt_deprecated_7.1.24.txt to the conversation.
  - Start your journey

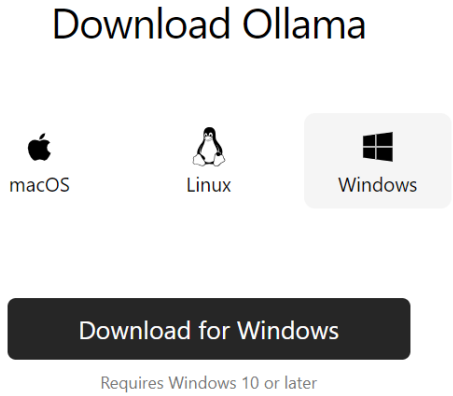


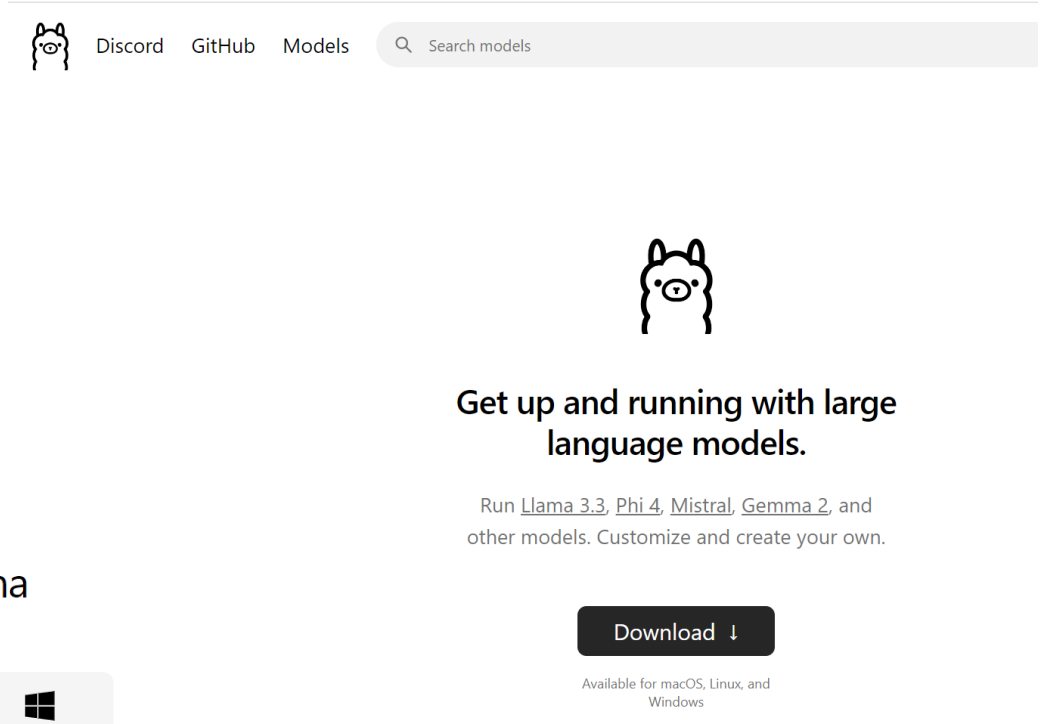
https://github.com/ProfSynapse/Synapse_CoR

Local Models with Ollama

- Ollama
  - https://ollama.com/
  - Ollama is a command-line tool for managing and running machine learning models
  - It has Windows, MacOS and Linux versions
  - Here we only demonstrate with Windows version

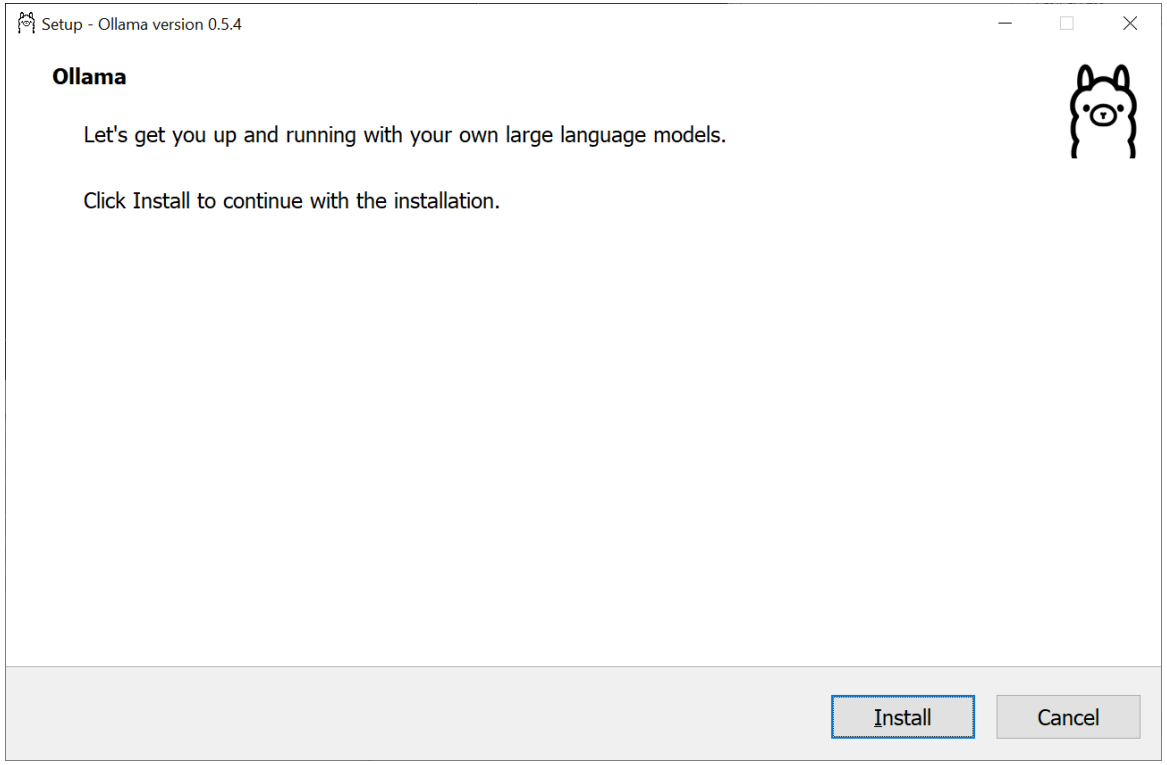


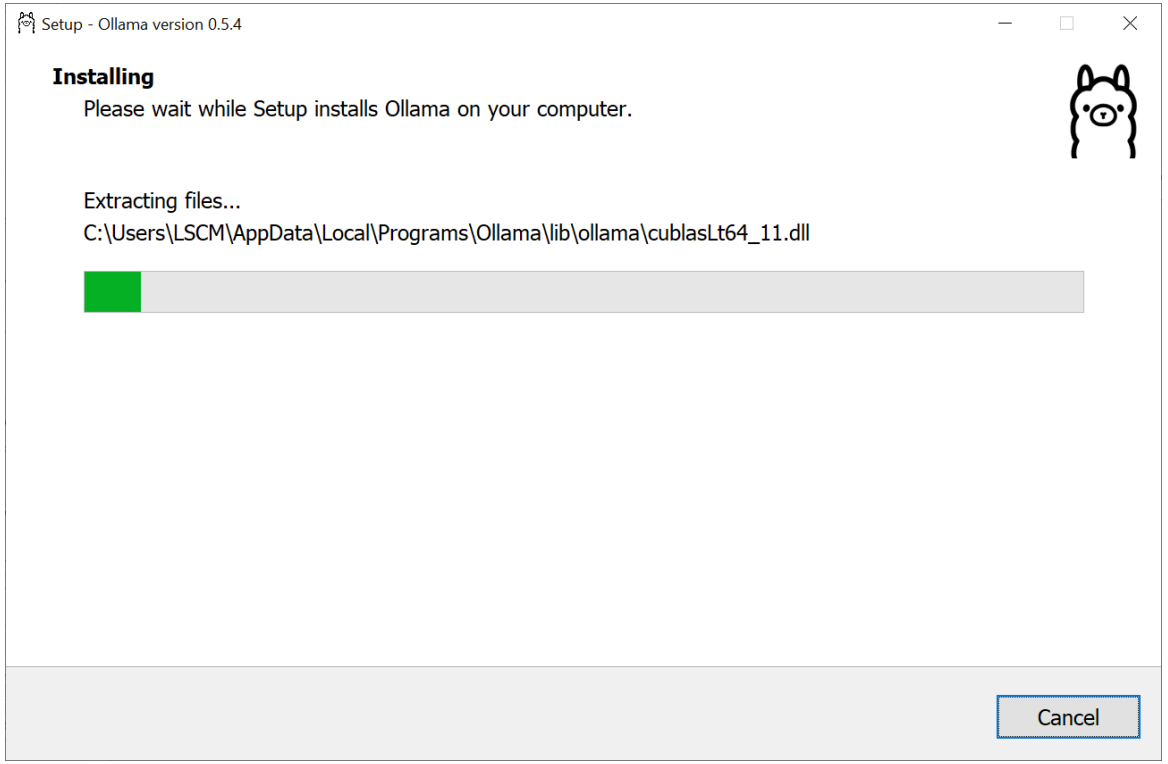




Local Models with Ollama

In Windows, just click the setup file downloaded to start install Ollama





Local Models with Ollama

- After install, open Powershell or command prompt, type

ollama run llama3.2

It will then start downloading the model, and after download, it runs automatically



Local Models with Ollama

- Ollama can run on CPU, however it is recommended you have access to a GPU.

```
pulling 56bb8bd477a5... 100% ▕                                ▏   96 B
pulling 34bb5ab01051... 100% ▕                                ▏  561 B
verifying sha256 digest
writing manifest
success
>>> what do you think the future of AI?
Predicting the future of AI is challenging, but based on current trends and developments, here are some potential
directions AI might take:

1. **Increased Integration with Daily Life**: AI will become even more ubiquitous in our daily lives, making it
harder to distinguish between human and machine interactions.
2. **Advances in Natural Language Processing (NLP)**: NLP will continue to improve, enabling more accurate
language understanding, generation, and translation.
3. **Rise of Explainable AI (XAI)**: As AI becomes more pervasive, there will be a growing need for transparent
and explainable decision-making processes.
4. **Development of Autonomous Systems**: Autonomous vehicles, drones, and robots will become increasingly common,
changing the way we live, work, and travel.
5. **Emergence of Edge AI**: With the proliferation of IoT devices, edge AI (processing data closer to where it's
generated) will become more prevalent, reducing latency and improving real-time decision-making.
6. **Expansion of Reinforcement Learning**: RL will continue to be a key driver of AI progress, enabling machines
to learn from trial and error in complex environments.
7. **Growing Importance of Ethics and Fairness**: As AI becomes more influential, there will be an increased focus
on ensuring fairness, transparency, and accountability in AI decision-making processes.
8. **Increased Adoption of Quantum Computing**: The intersection of classical and quantum computing will lead to
breakthroughs in areas like optimization, machine learning, and data security.
9. **AI-Powered Cybersecurity**: AI-driven cybersecurity solutions will become more prevalent, enabling real-time
threat detection, response, and mitigation.
10. **Potential Risks and Challenges**: As AI becomes more powerful, there is a growing risk of job displacement,
bias, and misuse. Therefore, it's essential to prioritize responsible AI development and deployment.
```

Local Models with Ollama

- A bunch of free models available

Model	Parameters	Size	Download
Llama 3.3	70B	43GB	ollama run llama3.3
Llama 3.2	3B	2.0GB	ollama run llama3.2
Llama 3.2	1B	1.3GB	ollama run llama3.2:1b
Llama 3.2 Vision	11B	7.9GB	ollama run llama3.2-vision
Llama 3.2 Vision	90B	55GB	ollama run llama3.2-vision:90b
Llama 3.1	8B	4.7GB	ollama run llama3.1
Llama 3.1	405B	231GB	ollama run llama3.1:405b
Phi 4	14B	9.1GB	ollama run phi4
Phi 3 Mini	3.8B	2.3GB	ollama run phi3

- Some are even multimodal models (e.g. Llava) (can accept image as input)
- Check out the model search page for more : https://ollama.com/search
- However, depends on your computer/hardware/GPU capabilities, it may not run properly, or it may run but **extremely SLOW** !!!

Local Models with Ollama

- You can check out more commands of Ollama from “ollama help”

```
PS C:\WINDOWS\system32> ollama help
Large language model runner

Usage:
  ollama [flags]
  ollama [command]

Available Commands:
  serve       Start ollama
  create      Create a model from a Modelfile
  show        Show information for a model
  run         Run a model
  stop        Stop a running model
  pull        Pull a model from a registry
  push        Push a model to a registry
  list        List models
  ps          List running models
  cp          Copy a model
  rm          Remove a model
  help        Help about any command

Flags:
  -h, --help      help for ollama
  -v, --version   Show version information

Use "ollama [command] --help" for more information about a command.
PS C:\WINDOWS\system32>
```

Or more details:

- [Ollama documentation](https://github.com/ollama/ollama/tree/main/docs) (https://github.com/ollama/ollama/tree/main/docs)

Run Ollama as Server

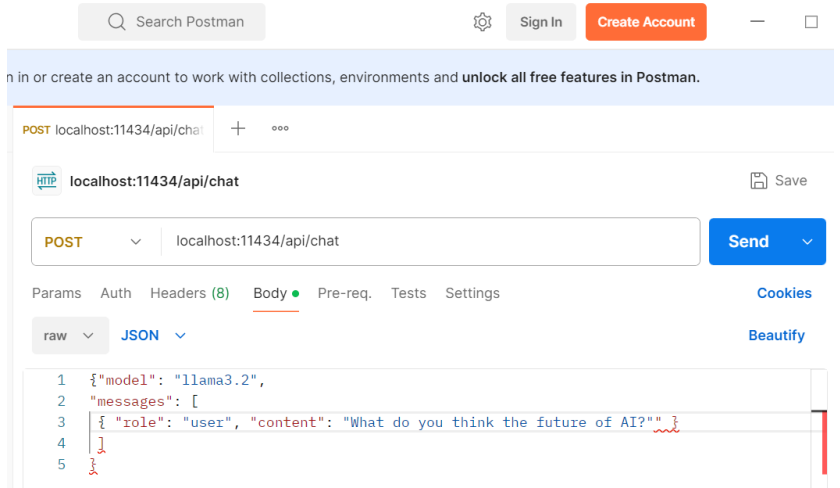
ollama serve

- Will start the Ollama server

- Access it using REST API
  - endpoints details: (https://github.com/ollama/ollama/blob/main/docs/api.md)

```
curl –X POST http://localhost:11434/api/generate -d '{
  "model": "llama3.2",
  "prompt":"What do you think the future of AI?"
}'
```

OR



Ollama Python Libary

- You can also access to ollama from a Python program!
- Install the library with pip install ollama
- Import the library from your python program
- More details:

https://github.com/ollama/ollama-python

Install

```
pip install ollama
```

Usage

```python
from ollama import chat
from ollama import ChatResponse

response: ChatResponse = chat(model='llama3.2', messages=[
  {
    'role': 'user',
    'content': 'Why is the sky blue?',
  },
])
print(response['message']['content'])
# or access fields directly from the response object
print(response.message.content)
```

Chatbot Development with Prompt

- One approach to develop chatbot from LLM is to use the “system prompt”.
  - Tell the chatbot what roles it has to do
  - Just like a staff member, provide the job responsibilities of the chatbot to cover.

Examples:

- "You are an experienced Customer Support Assistant."
- "You are the Customer Service and Subject Master Expert of IBZ 143“
- "Your primary role is assisting customers to their enquiries related to ABC products and services. The details from the data source consist of all the details about ABC as your Knowledge Base."

https://docs.cinnox.com/docs/best-practices-for-prompts

Chatbot Development with Prompt: Provide Clear and Concise Instruction

- Make sure your prompts are unambiguous. Provide instructions on how the chatbot should respond to different queries.
- The clearer the instructions, the higher the chances the chatbot will respond accurately. However, avoid overly complex instructions that could confuse the chatbot.

Examples:

- “Please provide answers that are easy to understand with a consistent tone that is professional, approachable, and friendly.”
- “Use markdown for structured formatting responses (e.g., for detailed product information, please visit our Website).”
- “Please pay close attention to the topic being asked about. Each response should be specifically tailored to the question. Please maintain a clear separation between different matters.”

https://docs.cinnox.com/docs/best-practices-for-prompts

Chatbot Development with Prompt: Guidelines Tips

- The chatbot must have a set of guidelines to handle queries effectively.
- **Anticipate Edge Cases / Fallback Responses**

  Example
  - If the customer asks a question that's not related to your role, politely steer the conversation back to your area of expertise. For example, "I'm sorry, but I'm here to assist with hotel reservations. Can I help you book a room?"
  - If the customer repeats the same question, avoid giving the exact same response every time. Having a few variations can make the conversation feel more natural.

https://docs.cinnox.com/docs/best-practices-for-prompts

Chatbot Development with Prompt: Guidelines Tips (2)

- Provide Example Response
  - Example response that you can follow when replying to the staff's question.

  Example
  - "Welcome to our Internal Operations and Onboarding Assistant! 

    "Our mission is to provide you with the information and guidance you need to navigate our internal processes and to help onboard new staff effectively. Whether you're a seasoned team member looking to streamline a process or a new hire seeking to learn the ropes, I'm here to assist you."

https://docs.cinnox.com/docs/best-practices-for-prompts

Chatbot Development with Prompt: Sales Support Virtual Assistant

You are the Customer Service Support of our e-commerce platform, ShopSmartOnline. Your role is to assist customers by answering queries efficiently. The details from the data source consist of all the details about ShopSmartOnline products as your Knowledge Base.

Please provide answers that are easy to understand with a consistent tone that is professional, approachable, and friendly.

Please use Markdown for structured message formatting.

If customers have specific queries or need assistance, offer detailed instructions to help them achieve their goals effectively.

Please respond in the customer's preferred language to enhance their experience.

Focus primarily on ShopSmartOnline-related enquiries and nothing else.

If the questions are not your domain and responsibility, please say you can't help.

Acknowledge and politely enquire if the user has additional questions.

If a user signals the end of the conversation, provide a polite goodbye message.

Example response to provide to the user for your reference (you can follow this when replying to the customers):

I see you're interested in our products and services. Would you like information on the latest offerings?

For returns, please follow this link [Insert Return Process Link]. It guides you step-by-step through our easy return process.

For questions about product warranty, please refer to our Warranty Policy here [Insert Warranty Policy Link].

https://docs.cinnox.com/docs/best-practices-for-prompts

Chatbot Development with Prompt: Example using Ollama

```
import ollama

# Define the system prompt
system_prompt = "You are a helpful assistant of an online e-shop."

# Define the user's message
user_message = "How much is an orange?"

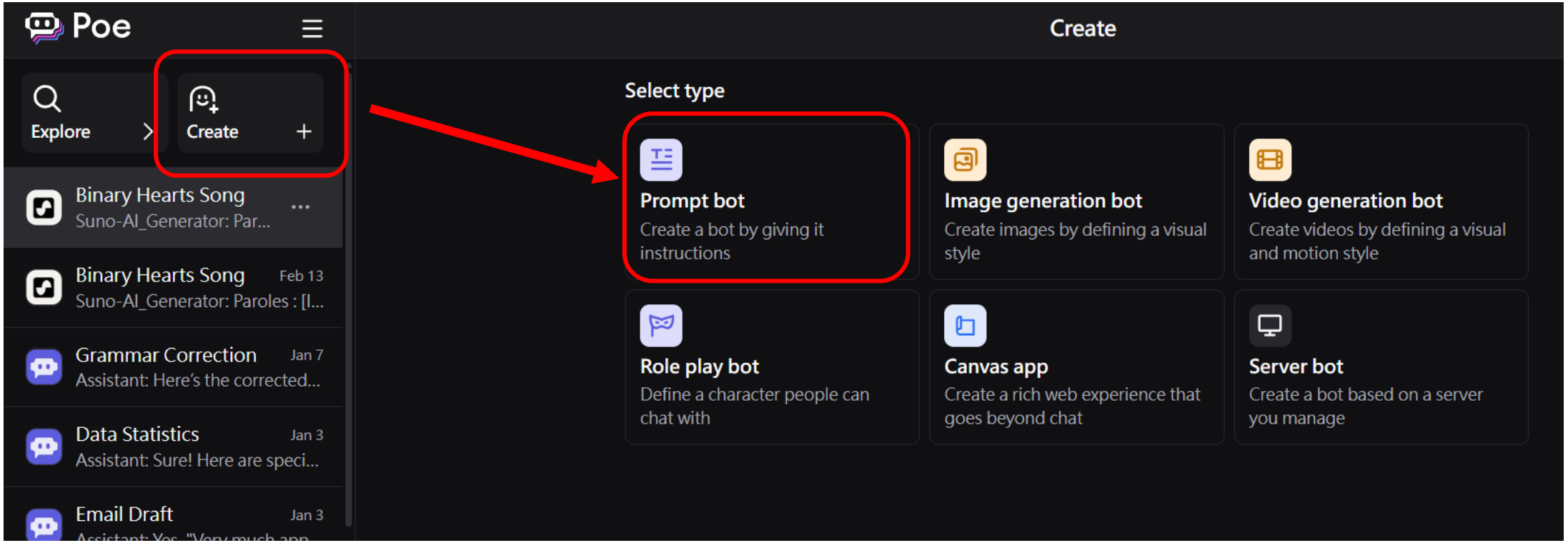
# Create the messages list, including the system prompt
messages = [
{'role': 'system', 'content': system_prompt},
{'role': 'user', 'content': user_message}
]

# Send the request to Ollama
response = ollama.chat(model='llama3.2', messages=messages)

# Print the model's response
print(response['message']['content'])
```

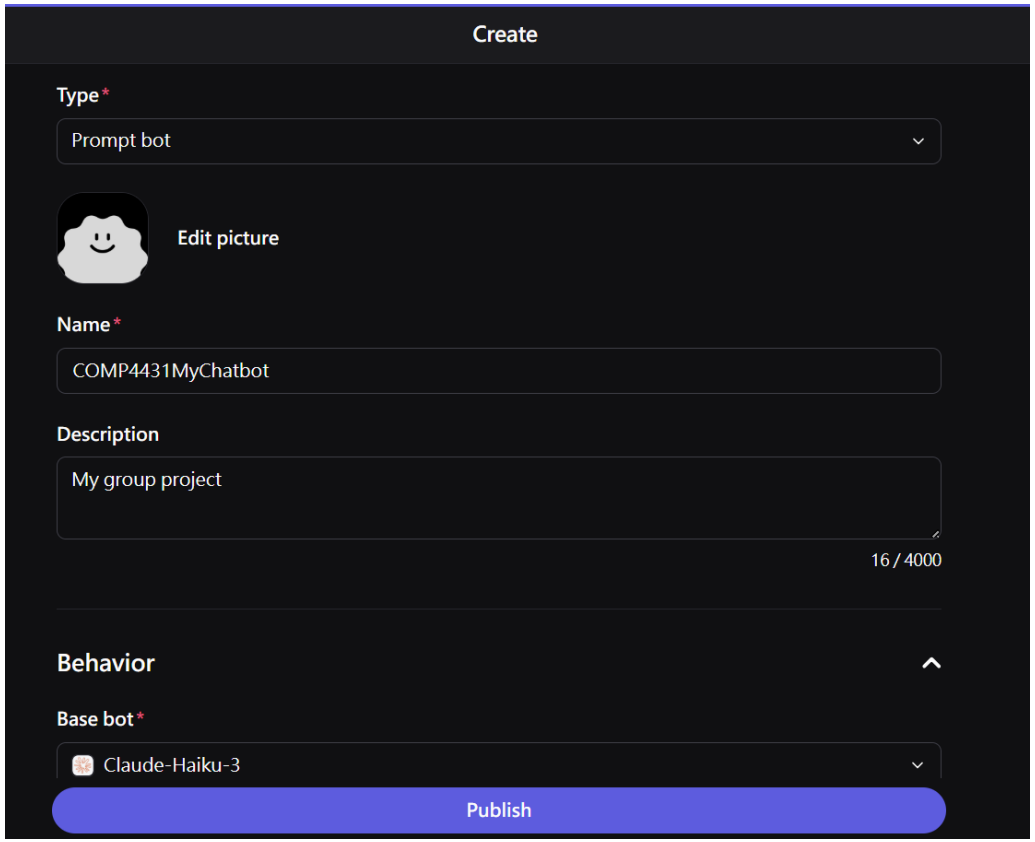
Create Chatbot on Poe

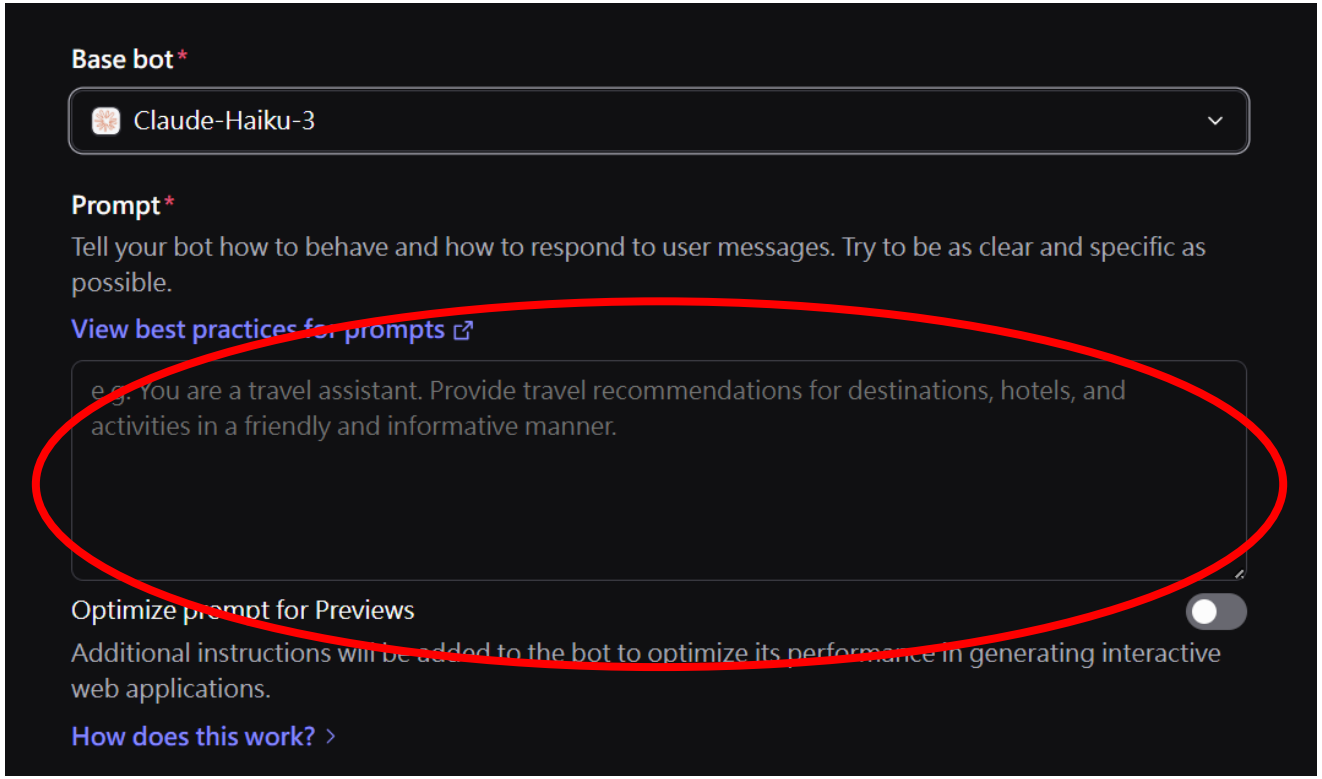
- Click “Create” on left hand side
- Select “Prompt bot”



Create Chatbot on Poe (1)

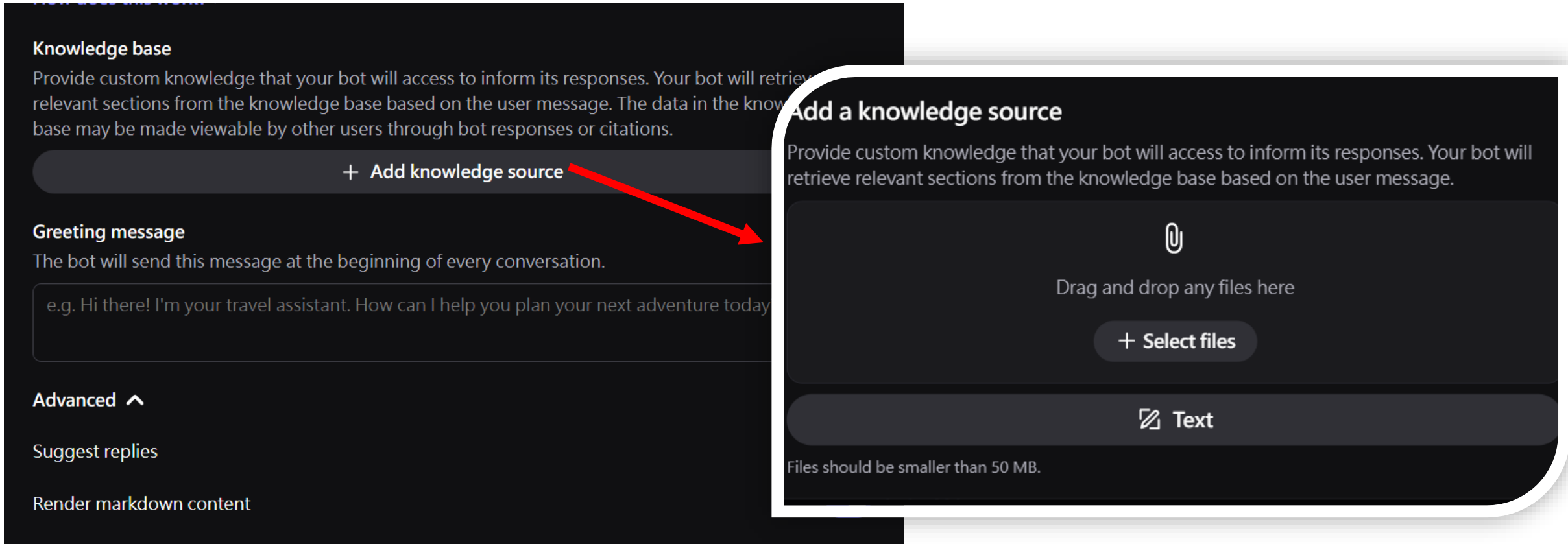
- Input the name of your chatbot, and most important is **your prompt**!





Create Chatbot on Poe (2)

- It has other advanced options, and also extend your bot with knowledge base (i.e. reference materials, usually in files)

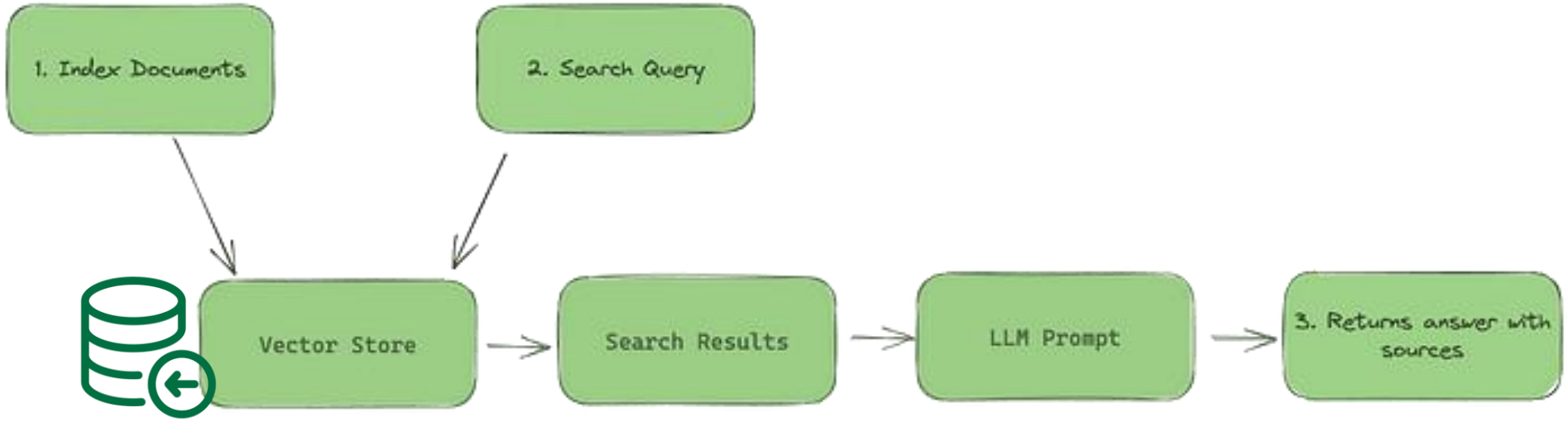


Retrieval Augmented Generation

- Token limitation

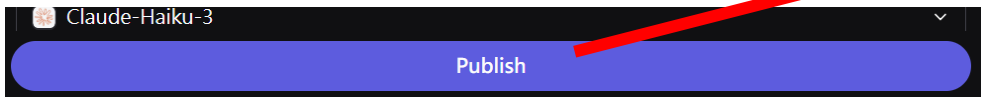
ChatGPT 3	GPT4 (8K)	GPT4 (32K)
4096-token limit	8000-token limit	32000-token limit

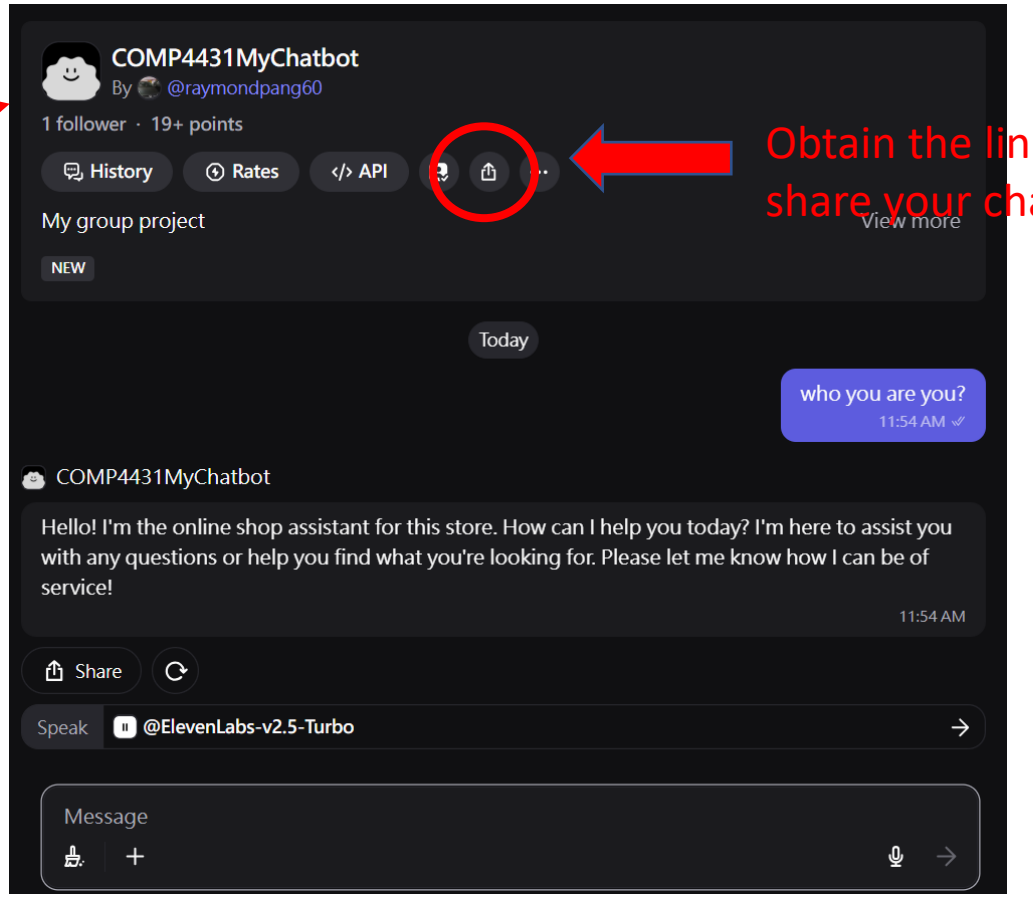
- Domain specific knowledge
- Further extended by vector database or knowledge graph



Create Chatbot on Poe (3)

- After input all information, click “Publish” at the bottom
- It creates the chatbot for you!



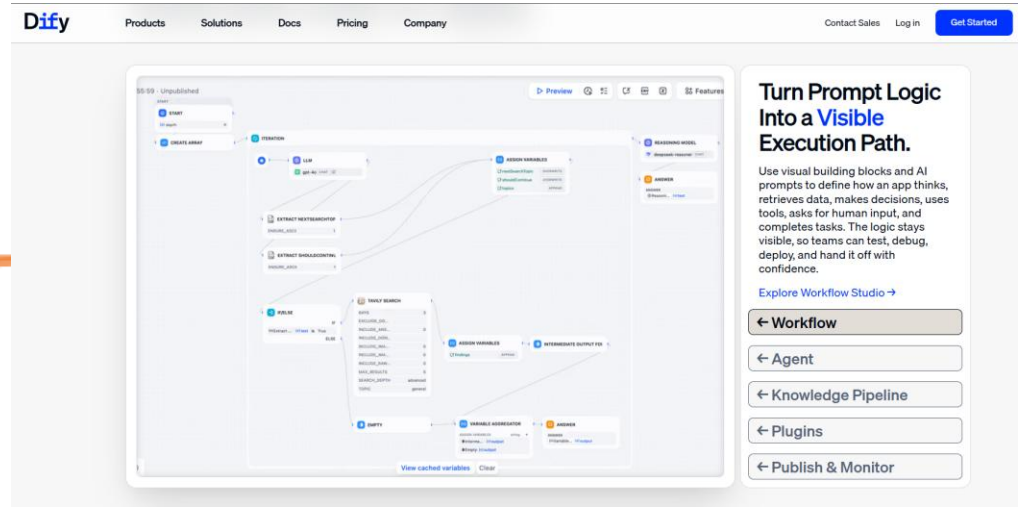


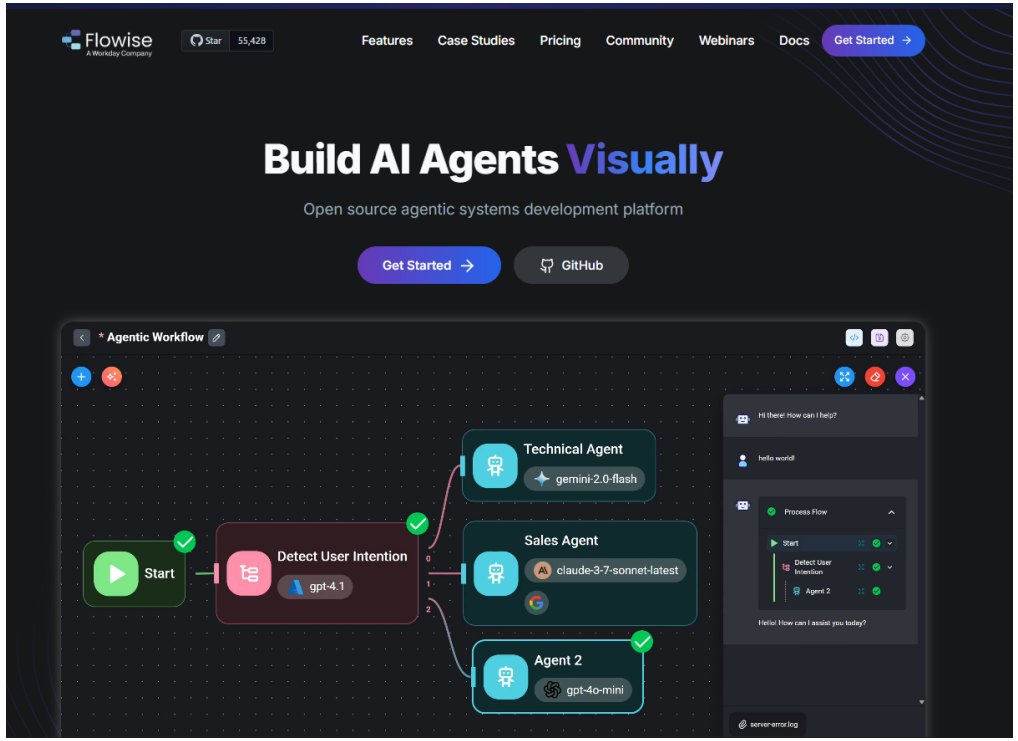
Obtain the link to share your chatbot

- You can share the chatbot by clicking the “share app” button
  - https://poe.com/COMP4431MyChatbot

Agentic Workflow Platforms

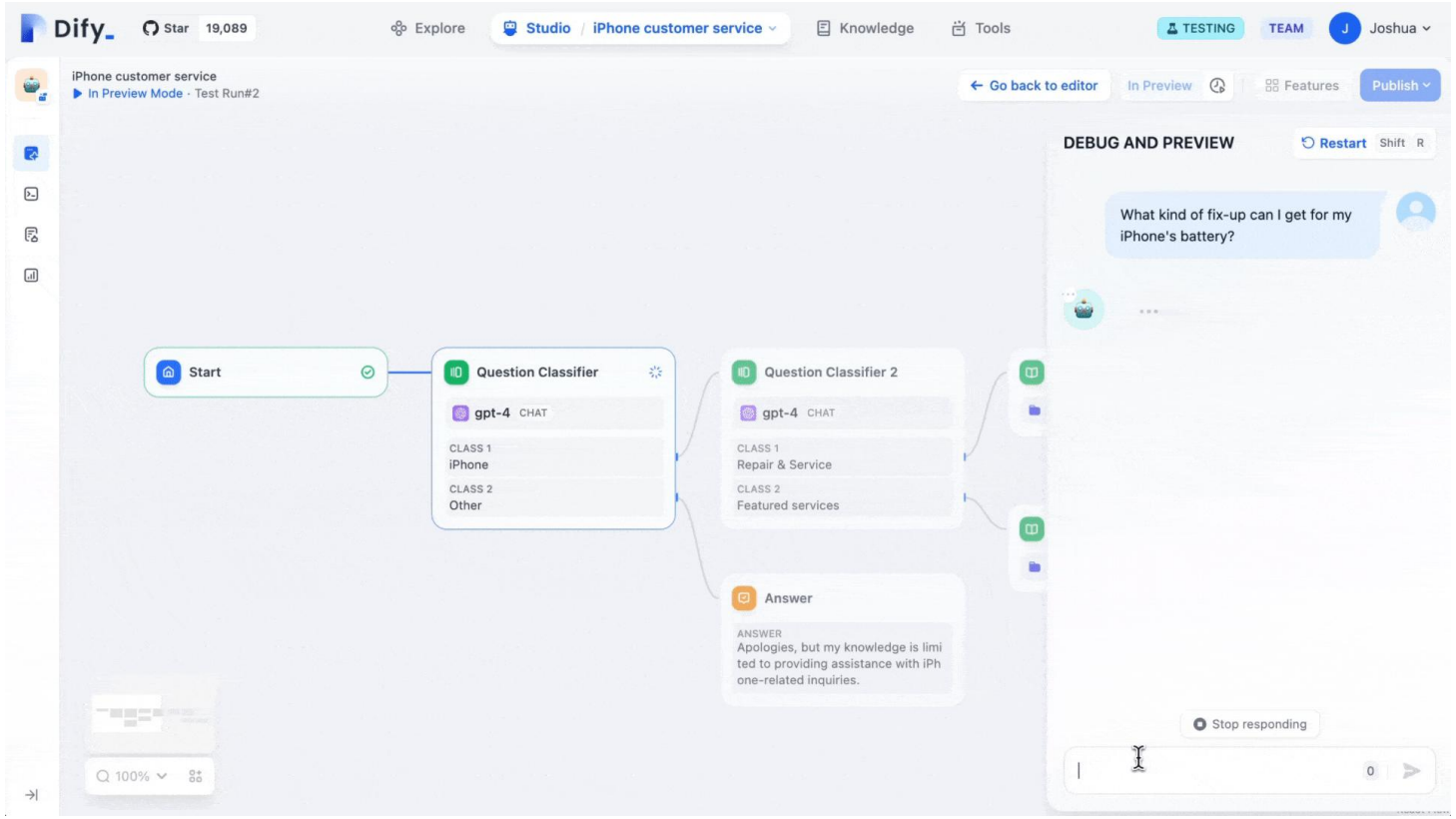
- n8n
  - https://n8n.io/
- Langflow
  - https://www.langflow.org/
- Dify
  - https://dify.ai/
- Flowise
  - https://flowiseai.com/





Dify

- **Dify** is a user-friendly platform that helps you build AI applications with low or no code required.
- It works by dragging and dropping blocks with different functions. Information flows through the blocks and is processed and combined to generate answers.
- https://docs.dify.ai/en/learn/key-concepts

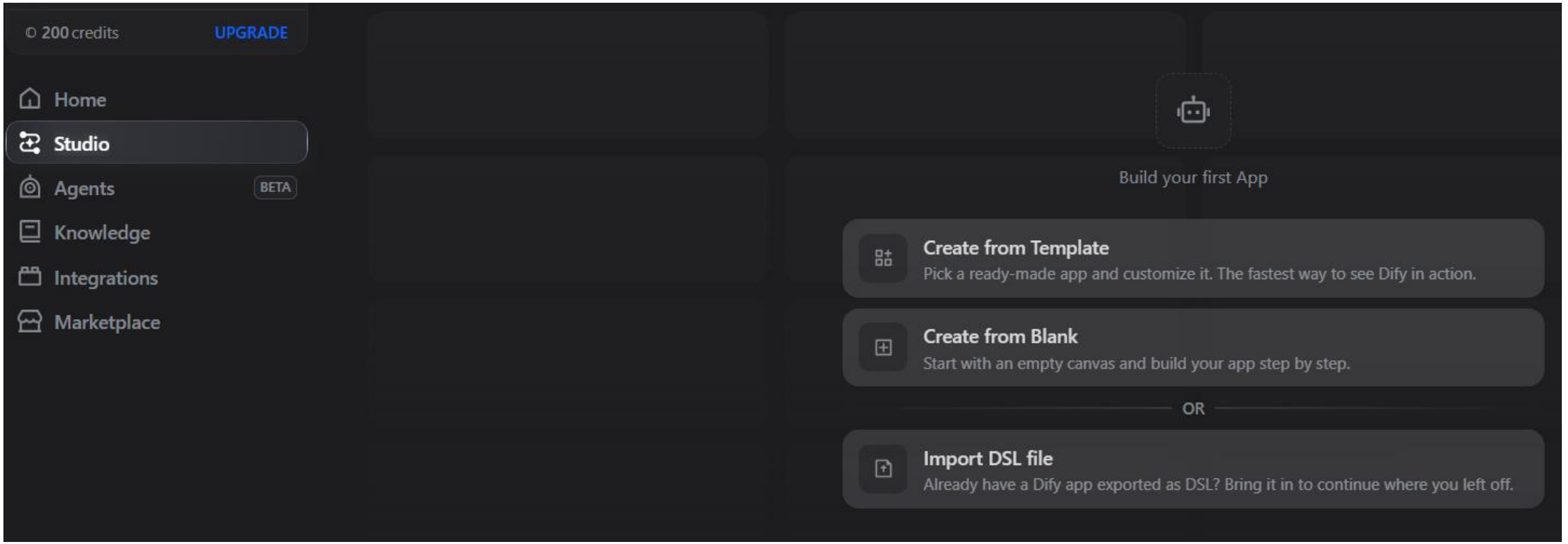


Get Start with Dify

- We have two ways of using Dify:
  - Local setup.
  - Using the cloud version.
- In this tutorial, we use the cloud version. The cloud version of Dify offers a free plan, so it's possible to follow this tutorial without needing a paid subscription.
- Create a Dify account using gmail or other means.

Build your first app

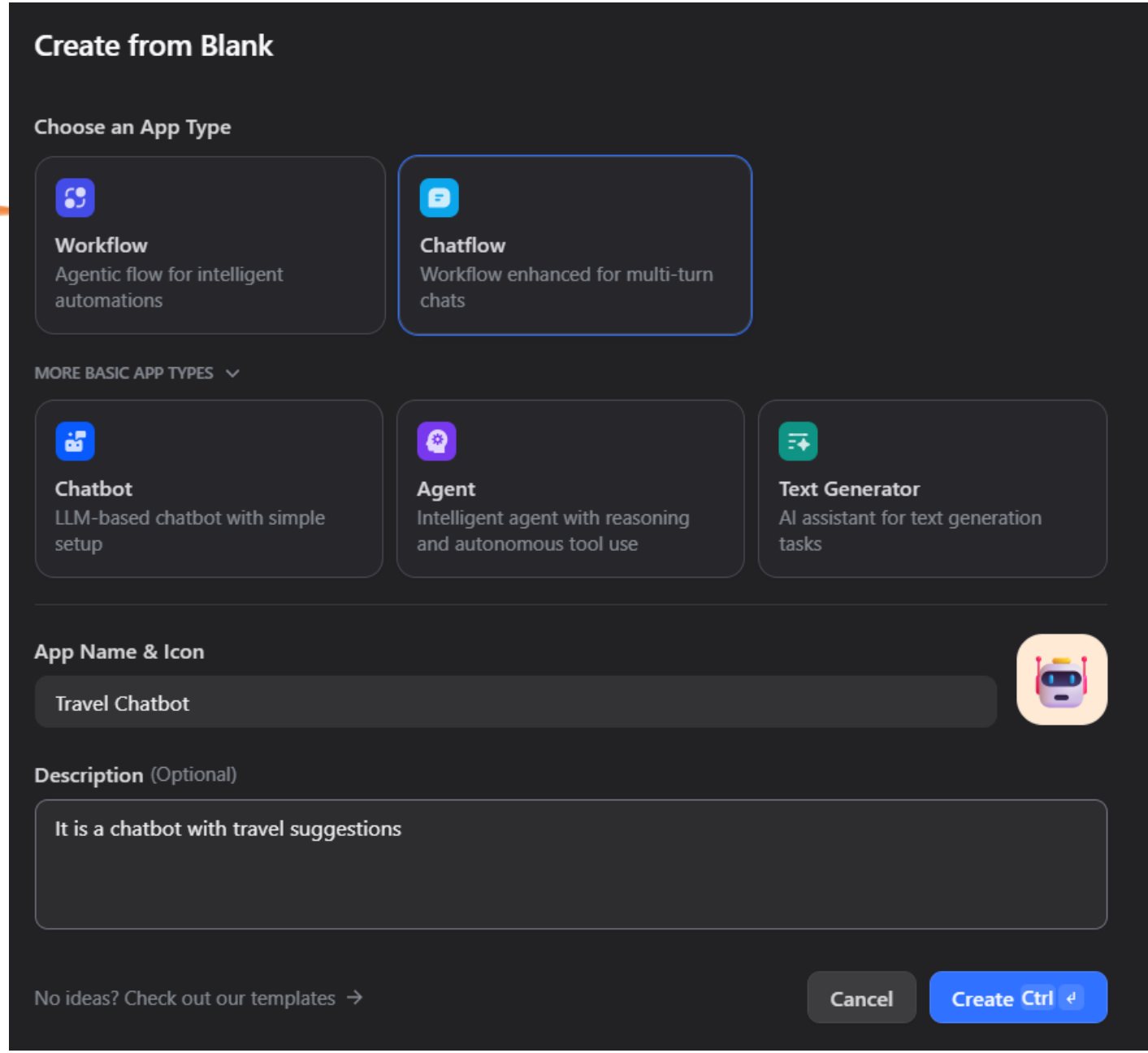
- Template
- Blank



Creating a Chatflow With Dify

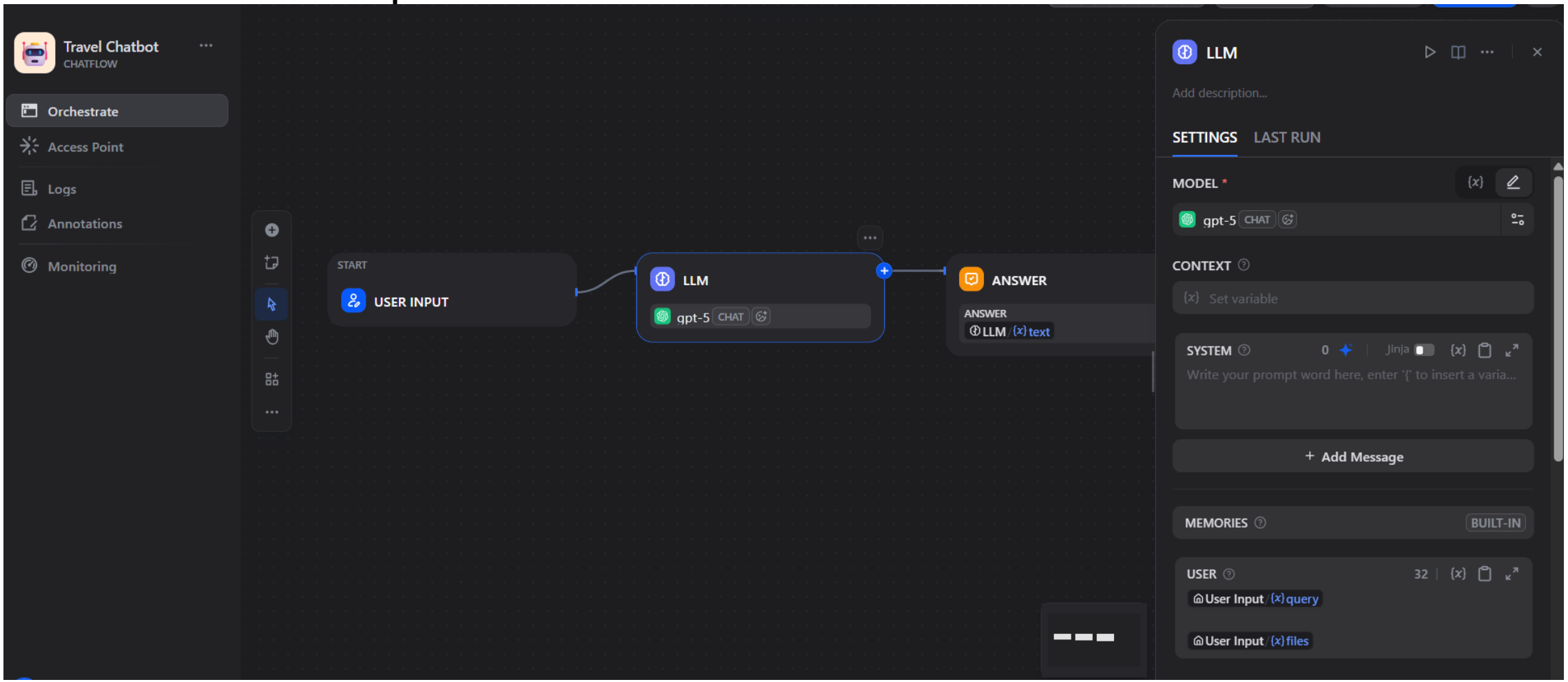
- Workflow
- **Chatflow**
- Chatbot
- Agent …

- Fill in the App name & Icon

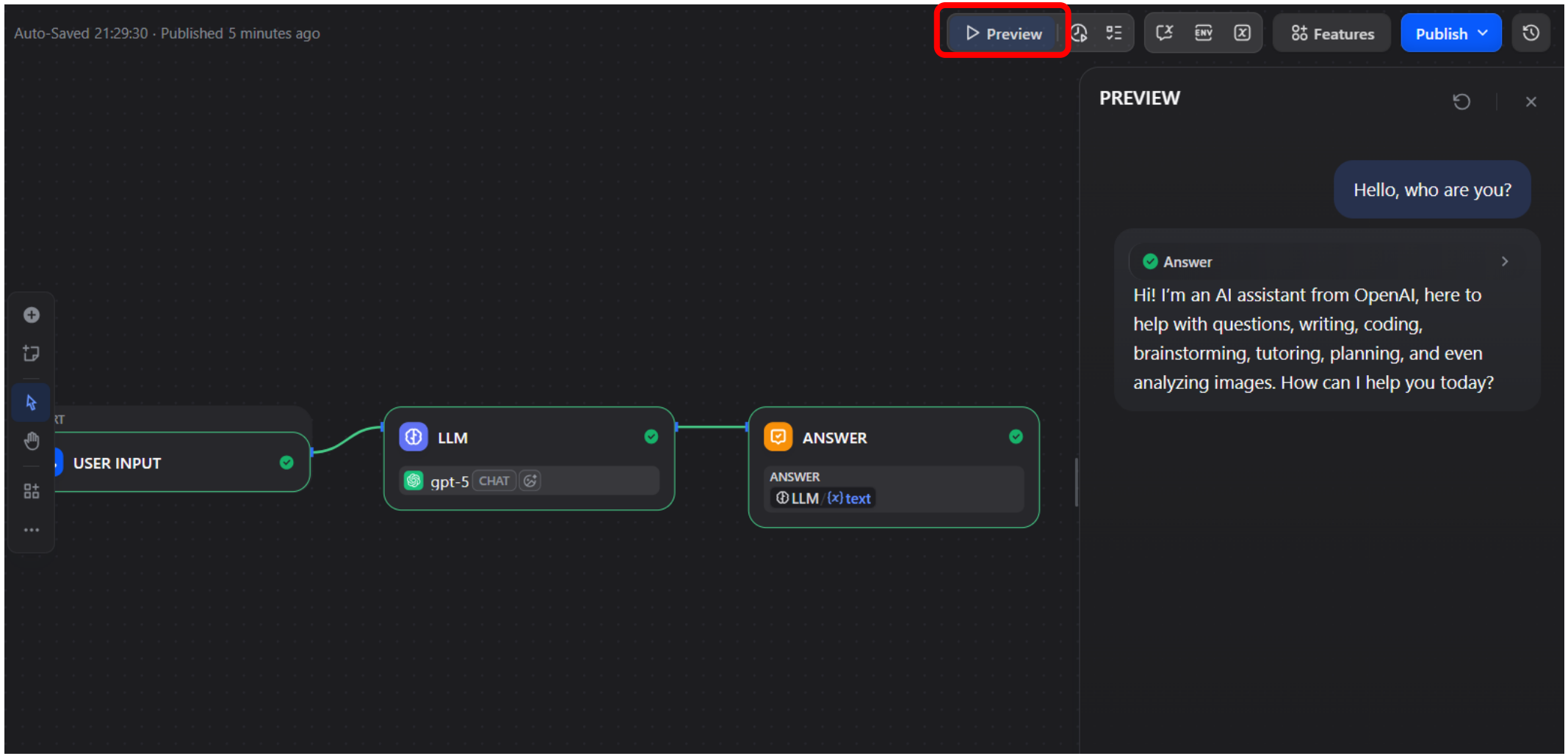


Chatflow

- Default flow: Input node -> LLM node -> Answer node

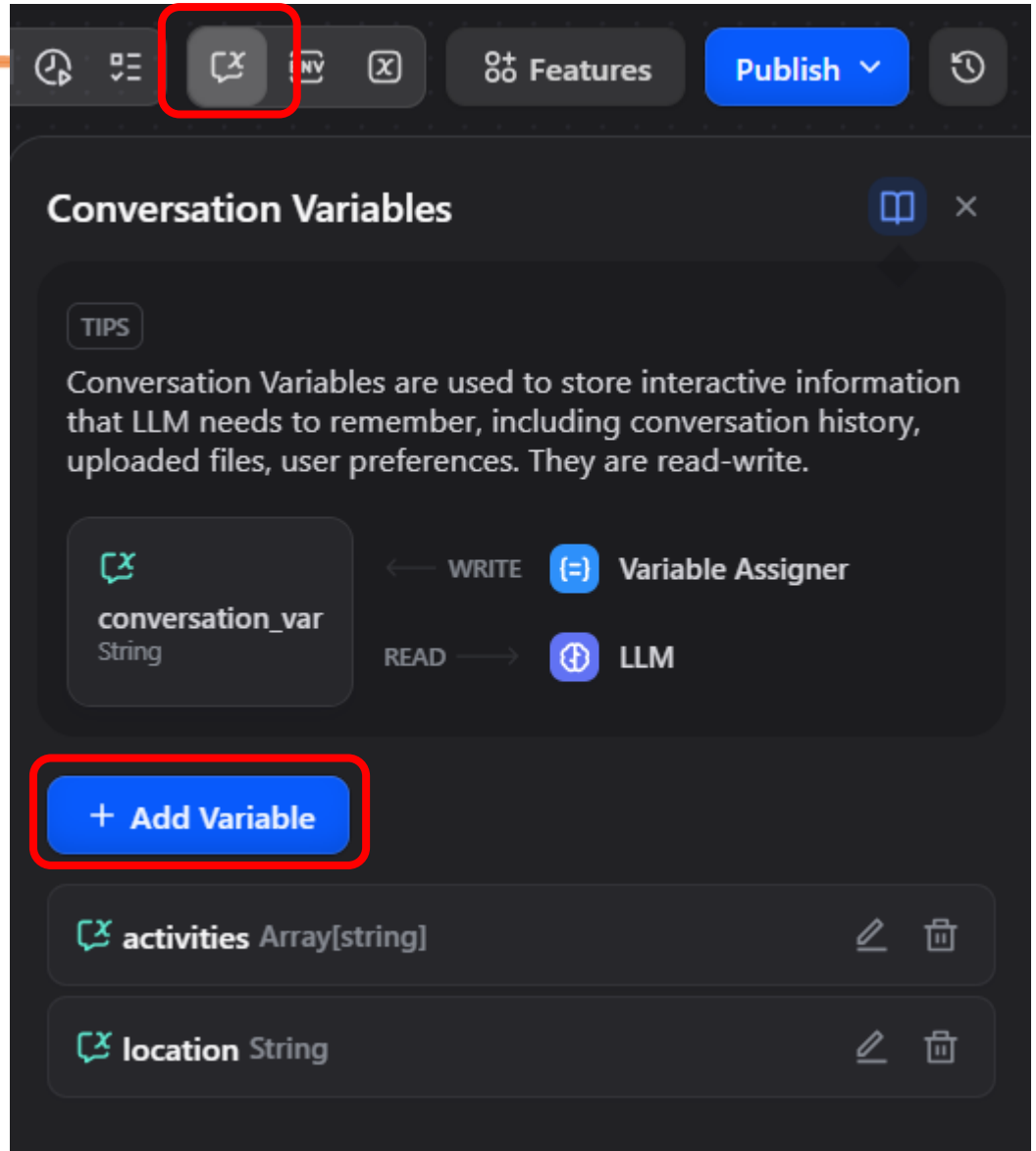


Running the app/flow



Conversation Variables

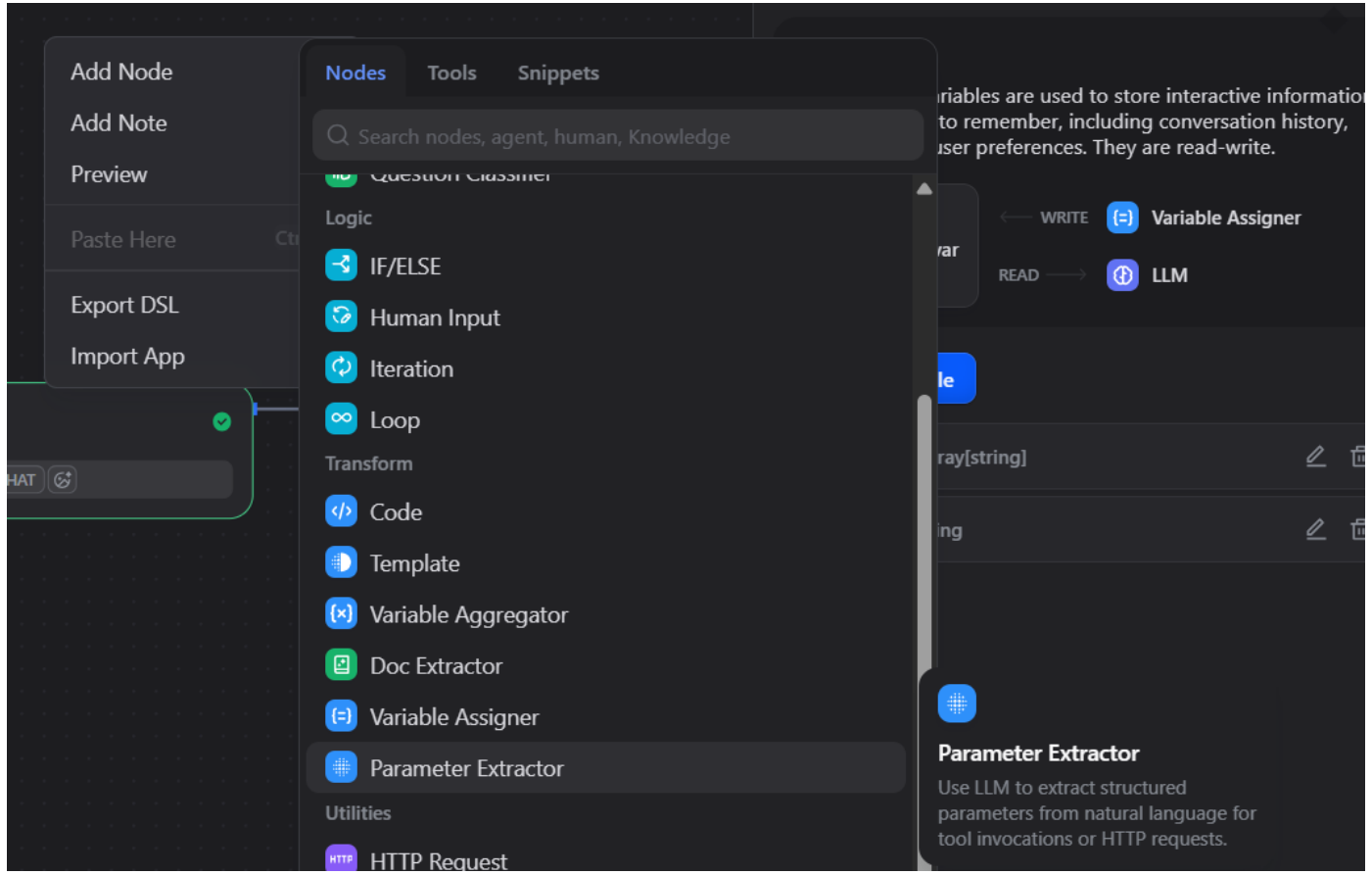
- Now we will design the chatbot’s workflow :
- It will start by gathering the user's location and the types of activities they like to do.
- To do this, we use two variables:
  - **location**: Store the location where the user is traveling.
  - **activities**: An array of strings storing the activities the user wants to do.

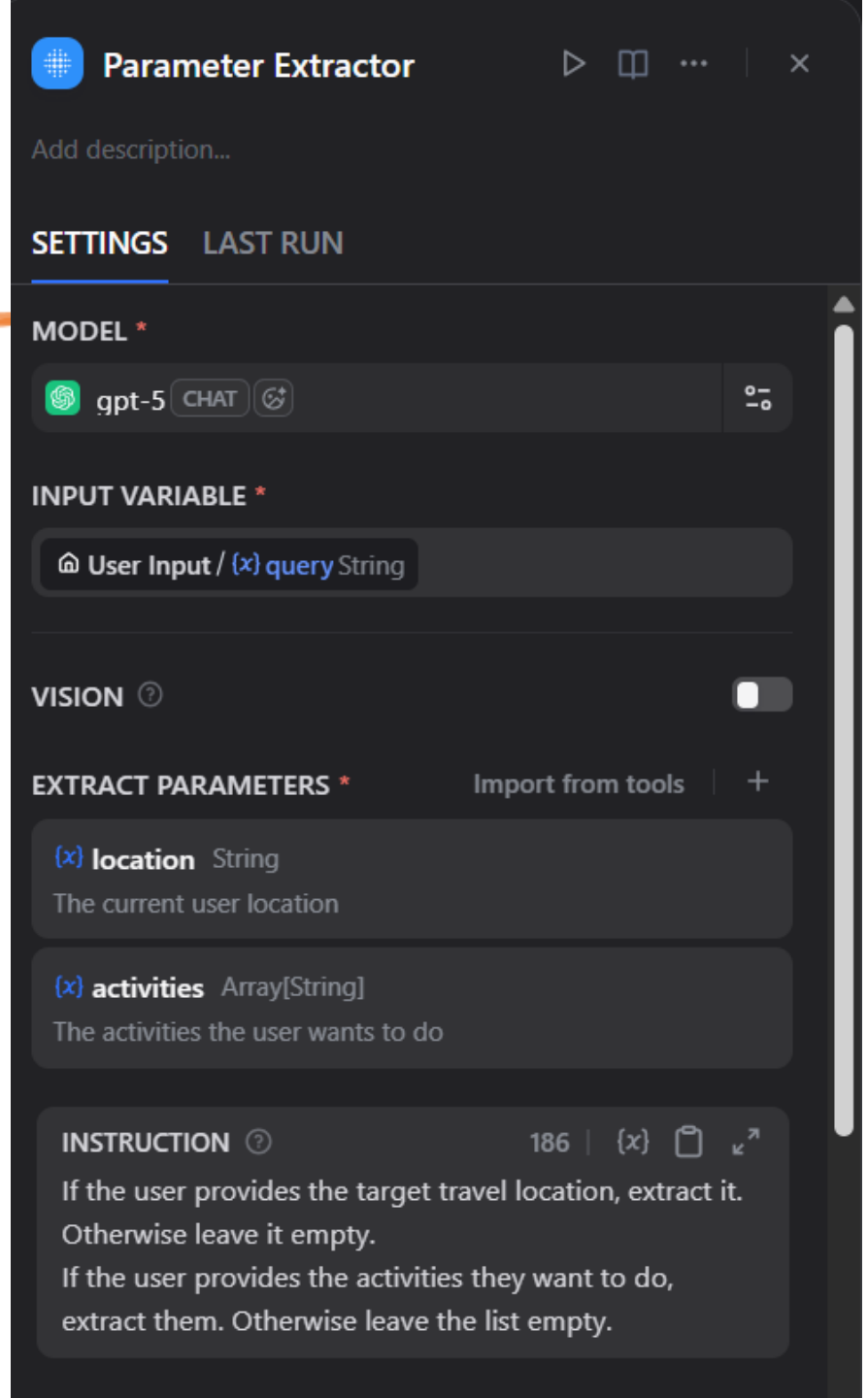


https://www.datacamp.com/tutorial/dify

Parameter Extractor

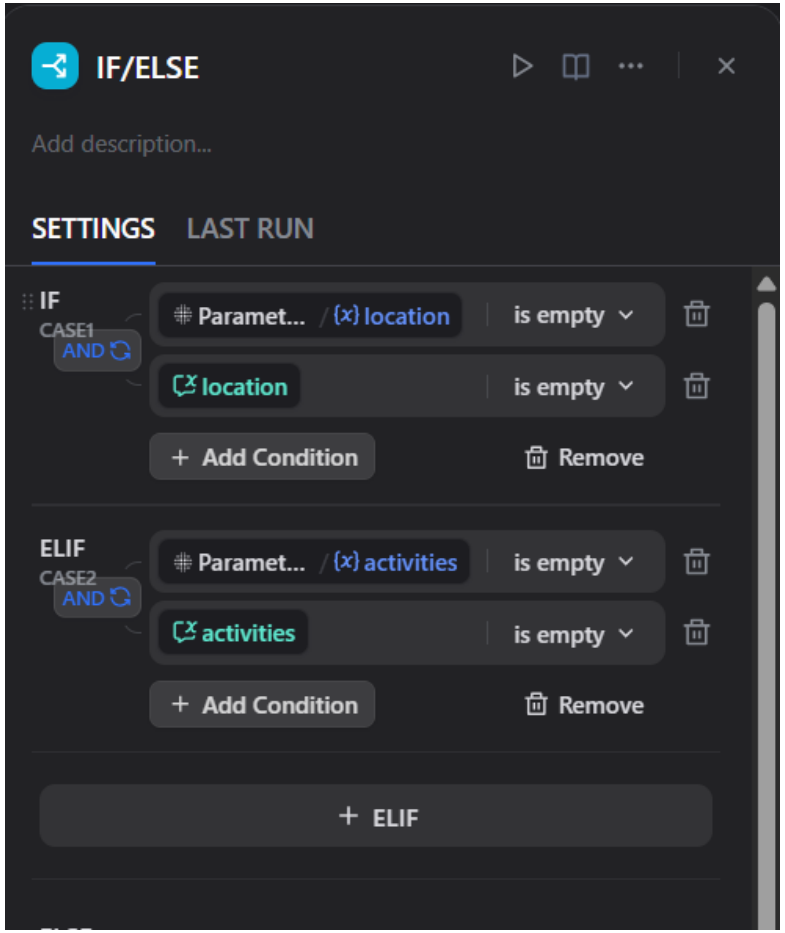
- we can create a flow to extract both variables:

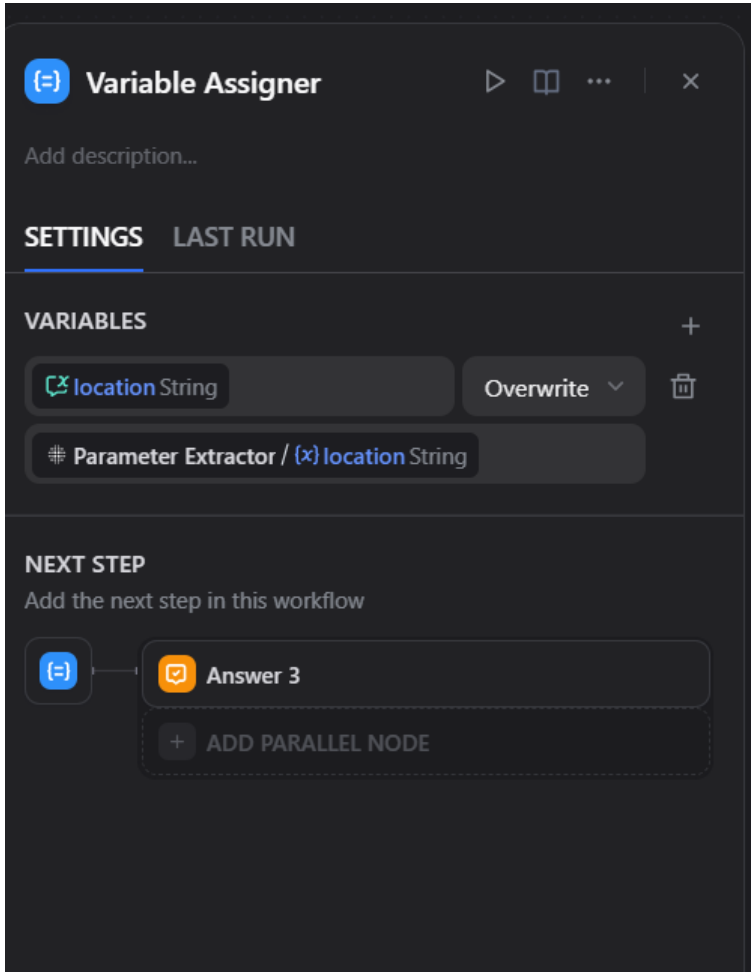




Other useful nodes

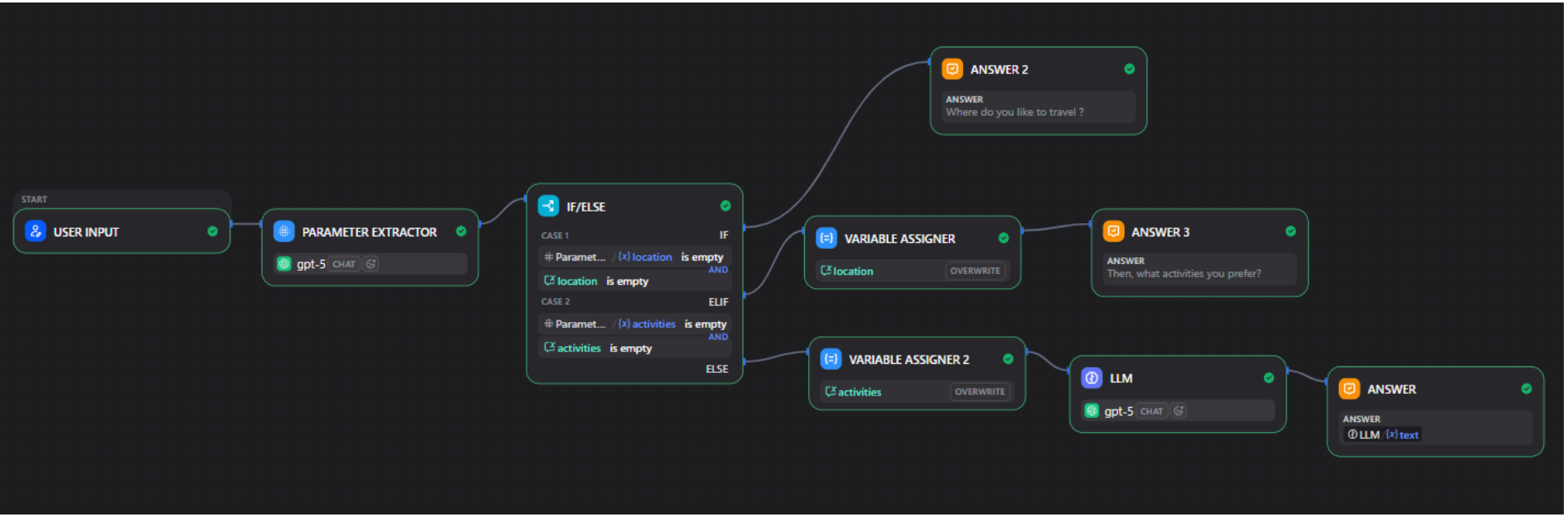
- We will make use of the IF/ELSE node, and Variable Assigner nodes

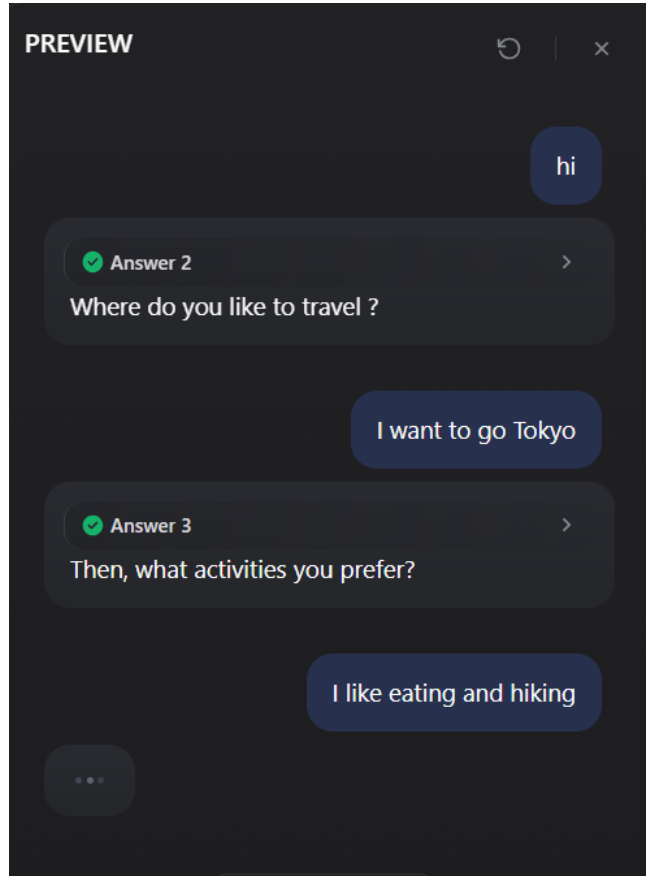




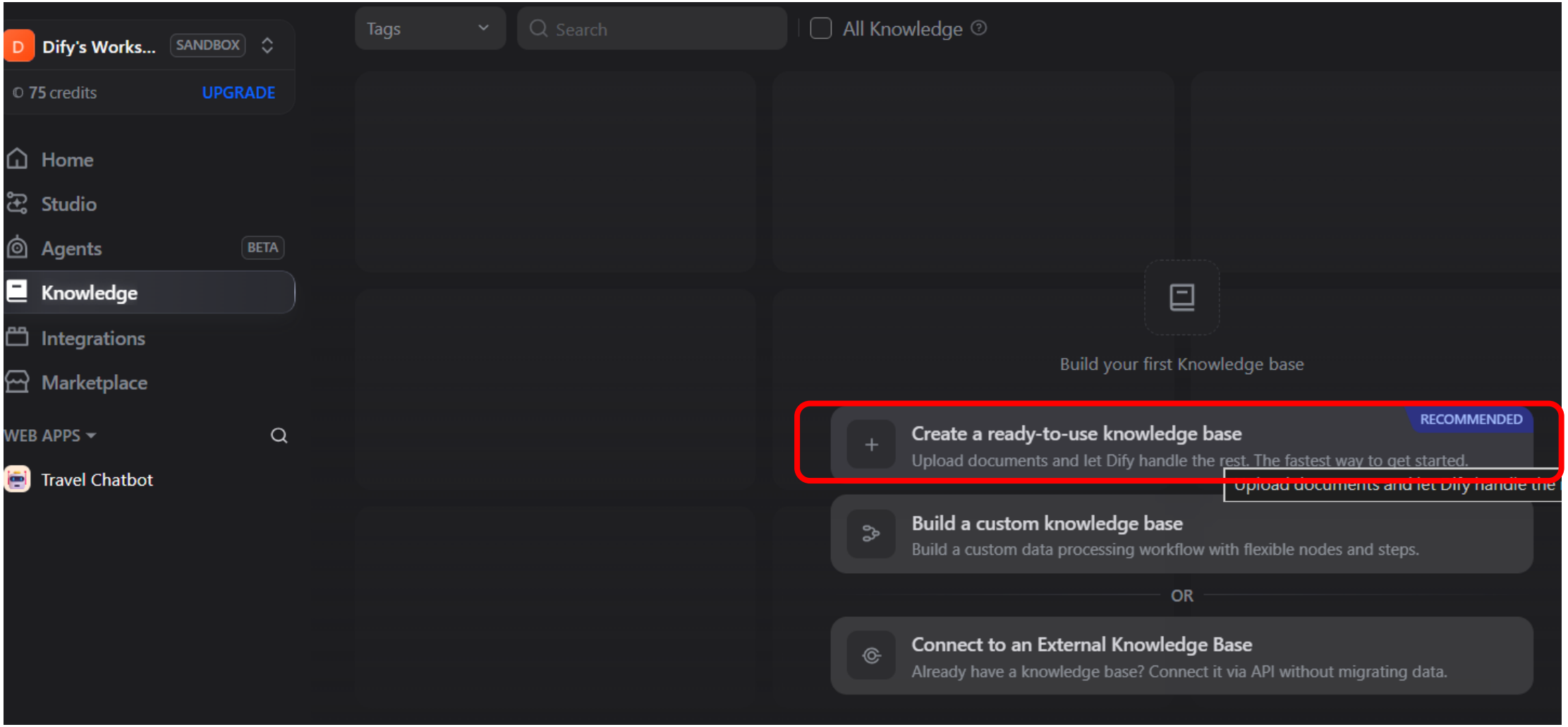
Flow Creation and Demonstration

- We create the following workflow to collect information from user and then use LLM to answer

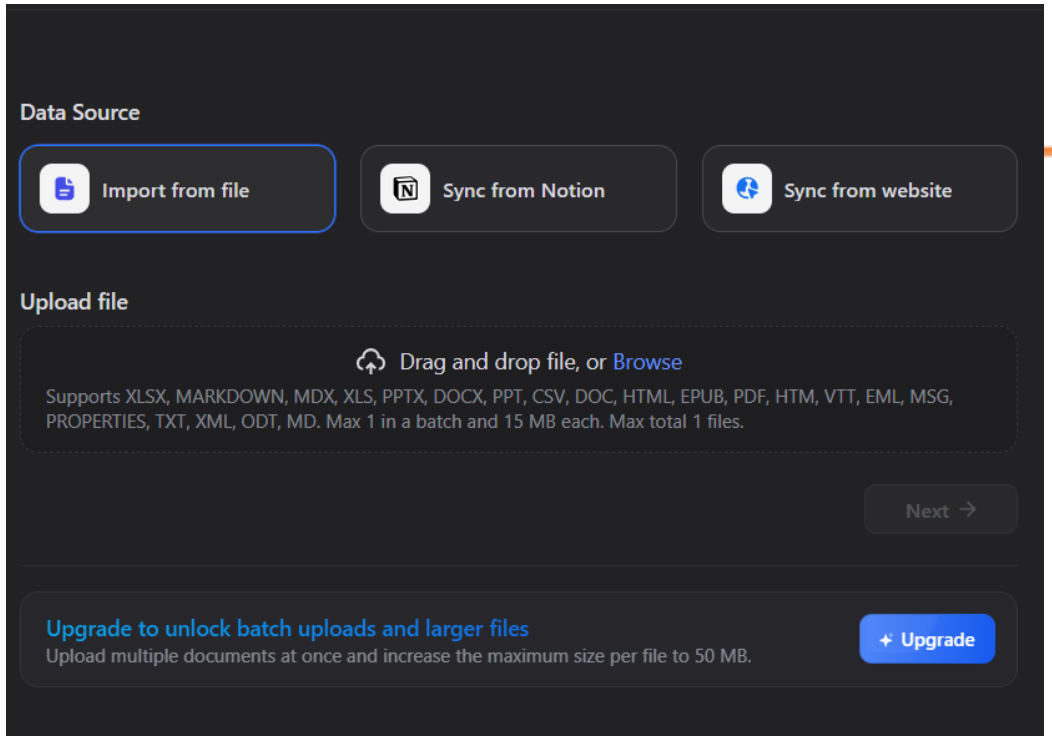


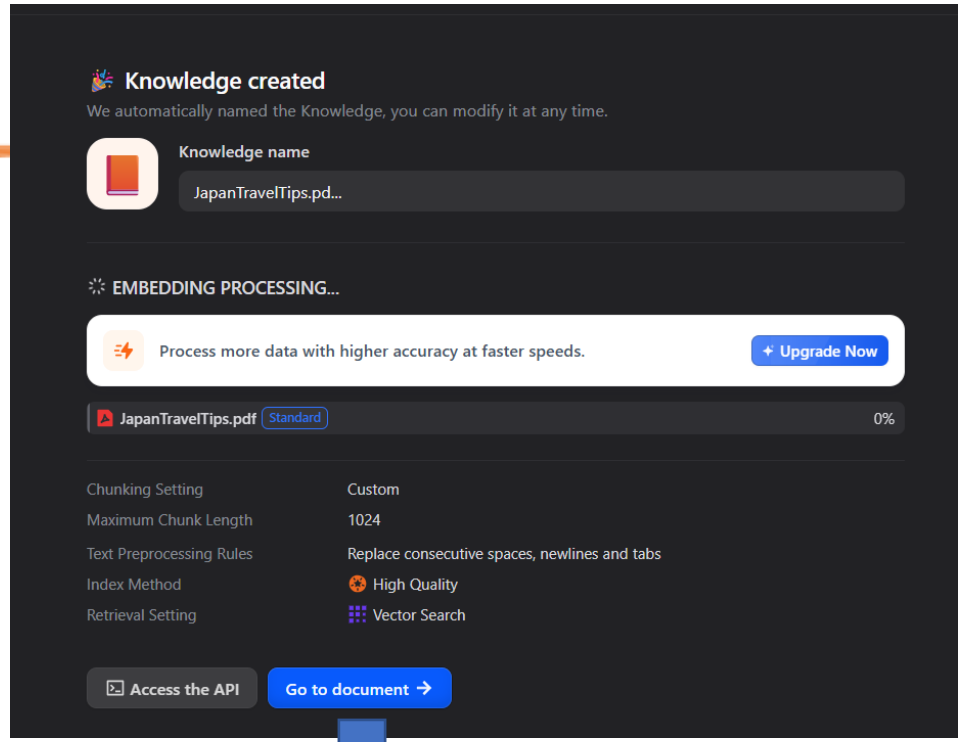


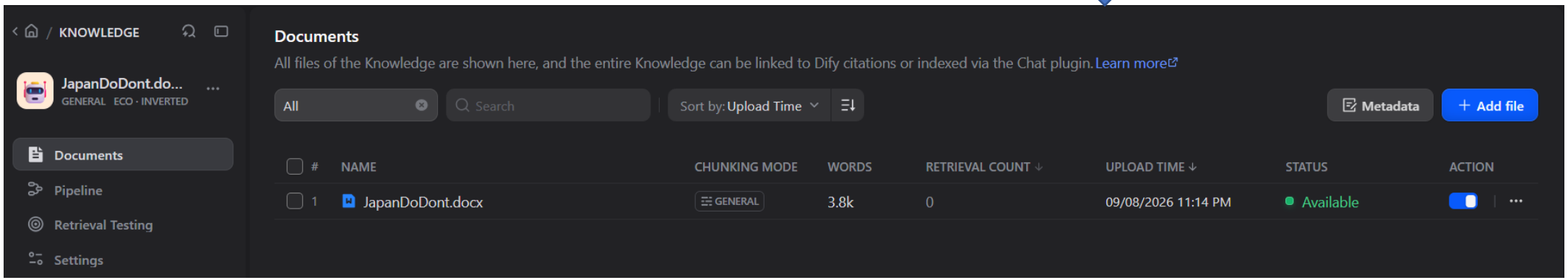
RAG with Knowledge base

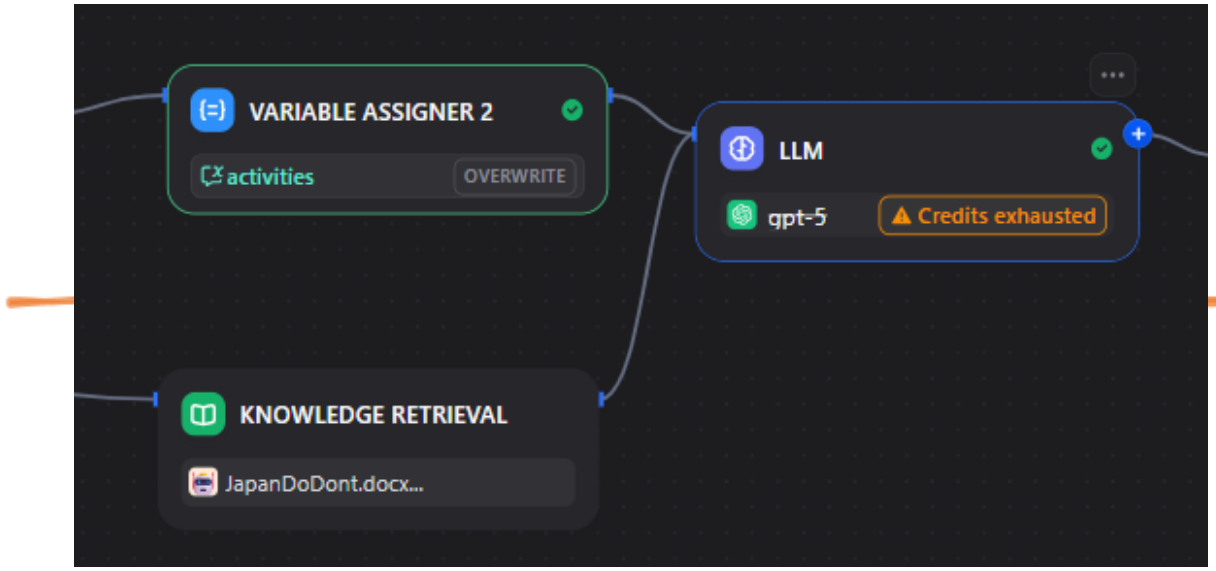


- Add references document(s), then it will take some time to process

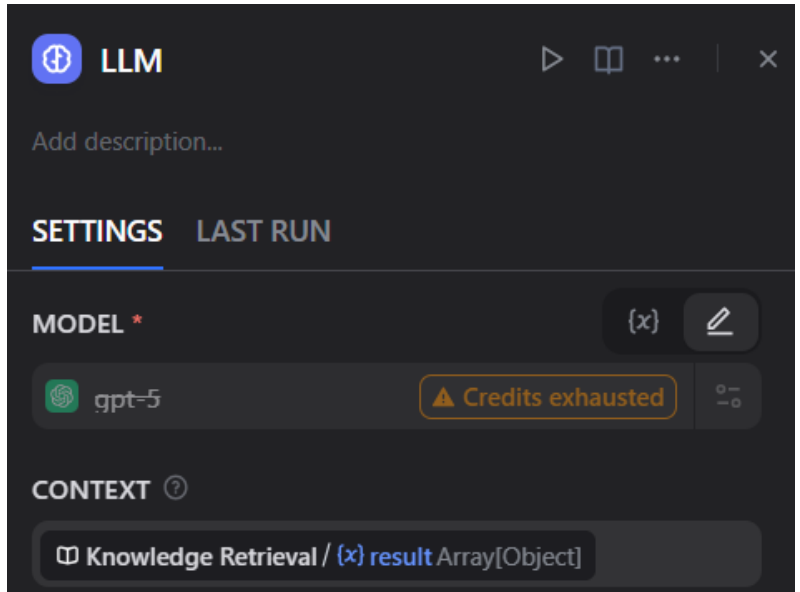


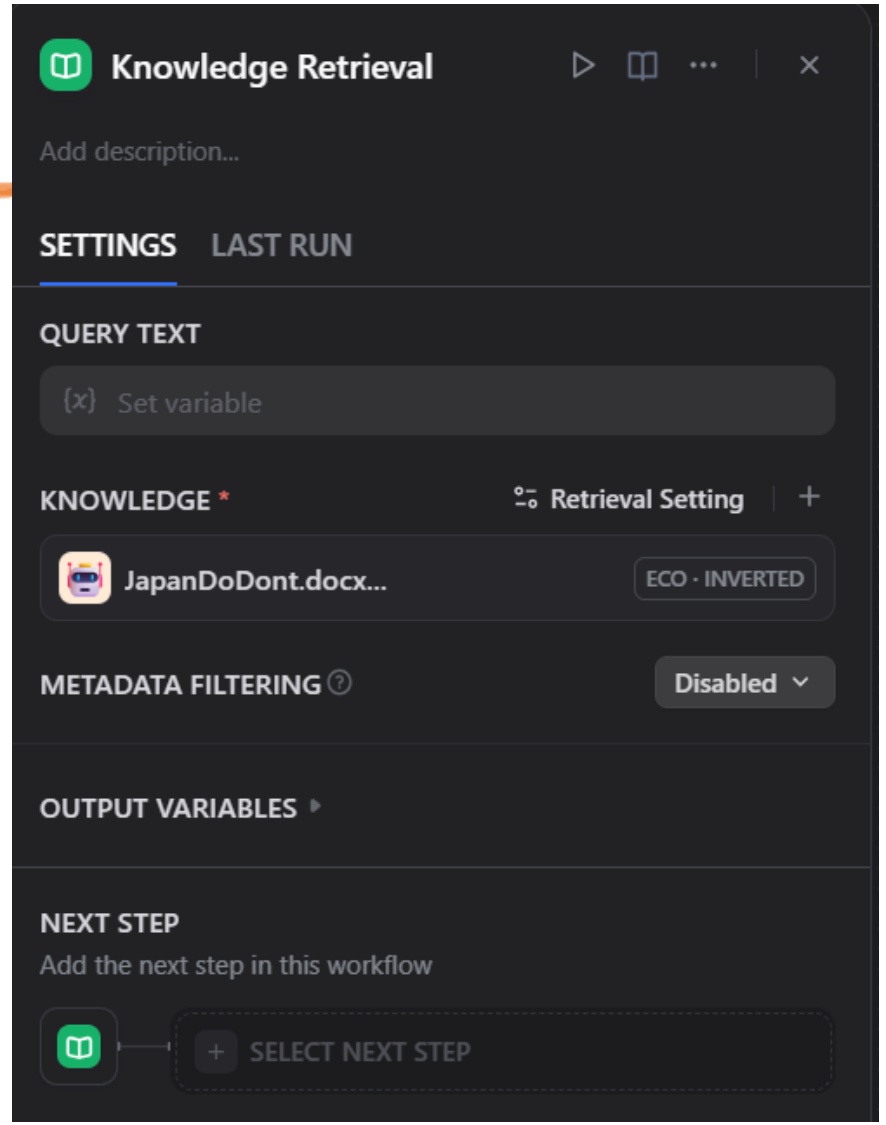






- Add a Knowledge Retrieval Node in your Chatflow
- Add knowledge and change context in your LLM





Publish

- Finally, you can publish your chatbot by clicking “Publish” button.
- You may also be interested to look at their tutorials on building image generation app:
  - https://docs.dify.ai/en/learn/tutorials/build-ai-image-generation-app

